# Financial Conditions and the Transmission of Monetary Policy Surprises

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**Abstract**

Does the estimated effect of a monetary policy surprise depend on how its financial consequences are measured? It does. Standard methods applied to surprises around Federal Reserve announcements give unemployment that falls after a tightening and prices that barely respond. These methods trace the surprise through one interest rate. A surprise also moves long yields, stock prices and the dollar, and by much more than an ordinary change in the two-year yield does. I measure these moves in the half hour around each announcement and carry them forward in a daily and a monthly model, with rates and financial conditions measured relative to the neutral rate. A tightening then raises unemployment and lowers the price level. Most of the change comes from the size of the initial financial movement, not from the dynamics. In the baseline model, a 1 percent lower core price level after two years comes with about 0.9 percentage-point years of additional unemployment. The unemployment result holds across specifications. The price result is clearest when the long yield and the dollar enter in proportion to their weights in the index.

**JEL Classification:** E43, E44, E52, C32

**Keywords:** monetary policy transmission, high-frequency identification, financial conditions, neutral interest rate, Phillips multiplier

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# 1 Introduction

Monetary policy is the main tool for stabilizing the economy, and its use rests on estimates of how policy moves unemployment and inflation. Those estimates are unreliable. In samples that start after the early 1980s, standard methods often give a tightening small effects on unemployment, or effects of the wrong sign (Ramey, 2016). Two concerns follow. Policy may have become less powerful (Boivin and Giannoni, 2006; Boivin, Kiley and Mishkin, 2010). Or inflation may respond so little to slack that lowering it requires a large and lasting rise in unemployment (Hazell et al., 2022; Del Negro et al., 2020). The cost of lowering inflation in unemployment is an old question (Okun, 1978; Tobin, 1980). The joint responses of unemployment and prices to a monetary surprise measure that cost at a stated horizon, along the path policy takes to lower prices.

I ask whether these responses depend on how the financial consequences of a surprise are measured. Policymakers think they do. In November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher longer-term bond yields, among other factors" (Powell, 2023). A surprise that raises the two-year yield also raises longer yields and the dollar and lowers stock prices, and it does so by much more than an ordinary rise in the two-year yield does. A model that traces the surprise through the two-year yield alone misses most of this movement.

The answer is that the measurement matters. I use the policy-news surprise of Nakamura and Steinsson (2018) around 252 FOMC announcements from 1996 to 2026 and change only how its effects are carried through the economy. Carried through the nominal two-year yield alone, the surprises give the result of the standard methods: unemployment falls and prices do not respond. Carried through the yield and a broad financial conditions index, with both measured relative to the neutral rate $r^*$, a tightening raises unemployment and lowers the price level (Figure 1). After two years the unemployment response rises from $-0.07$ to 0.50 percentage point and the fall in the price level deepens from 0.44 to 0.81 percent. Adding the index shifts the unemployment response by a similar amount when the yield is nominal or real. Most of the shift comes from the size of the initial financial movement, not from the dynamics that carry it forward (Section 3.4). In the baseline model, a 1 percent lower core price level after two years comes with about 0.9 percentage-point years of additional unemployment, one point of unemployment for one year.

The approach has three features. First, I measure the effect of each surprise in a short window around the announcement, when unemployment and inflation cannot move, and carry it forward in a daily and a monthly model estimated on ordinary data. As in Jacobson, Matthes and Walker (2026), the model takes into account when each announcement falls within its month. Second, financial conditions enter as a whole (Boivin, Kiley and Mishkin, 2010). The financial conditions index (FCI) combines the 10-year yield, equity valuation and the real dollar, weighted by their responses to policy news. Third, I measure the yield and the index relative to $r^*$, the rate that neither speeds up nor slows down the economy. The neutral rate fell from the 1990s to the 2010s (Laubach and Williams, 2003; Holston, Laubach and Williams, 2017, 2023; Hamilton et al., 2016; Bauer and Rudebusch, 2020; Kiley, 2020*b*), and rates trended down with it while activity trended

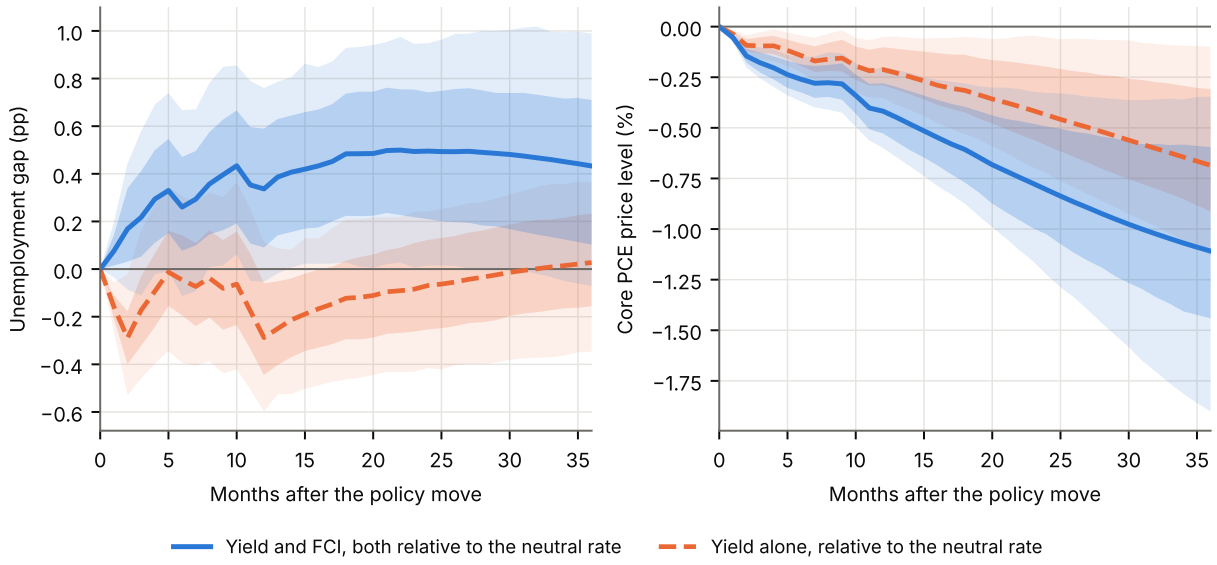



Figure 1: Responses to a Policy Move That Raises the Two-Year Yield 1 Percentage Point

*Notes:* The move occurs in the window around the announcement. Blue: the surprise carried through the two-year yield and the financial conditions index, both relative to the neutral rate. Orange dashed: carried through the two-year yield alone, relative to the neutral rate. Same surprise, normalization, sample (1996–2026), 12 monthly lags and bootstrap draws. Shaded bands: 68 and 90 percent intervals.

up. A model that links the level of rates to activity partly picks up these opposite trends. The adjustment should matter most for prices, which respond to how long conditions stay tight.

Two sections test the results. Section 3.6 asks whether the three parts of the index should enter in proportion to their weights. Section 4 varies the sample, the measure of $r^*$, the trend, credit conditions and the surprise.

## 1.1 Connection to the Literature

**Surprises and their effects.** Interest-rate surprises around FOMC announcements were meant to settle the measurement problem. The decision precedes the announcement, so market moves in a narrow window around it cannot have caused it (Kuttner, 2001; Gürkaynak, Sack and Swanson, 2005). They measure the effects of policy on asset prices well (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021), but for unemployment and inflation the old puzzles often return. Four explanations come with fixes. Announcements reveal the Fed's view of the economy, and separating it gives larger price effects (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). Part of each surprise is predictable, and removing it sharpens the estimates (Bauer and Swanson, 2023*b*,*a*). Markets react in minutes while unemployment and inflation are monthly averages, and measuring inflation daily removes most of the wrong sign (Jacobson, Matthes and Walker, 2026). And credit costs, stock prices and the dollar carry part of the effect (Gertler and Karadi, 2015; Caldara and

Herbst, 2019; Kroencke, Schmeling and Schrimpf, 2021; Bauer, Bernanke and Milstein, 2023; Boehm and Kroner, 2024; Bauer, Schmeling and Schrimpf, 2026; Boehm, Kim and Kroner, 2026). Most of this work traces the surprise through a single nominal interest rate.

**What is new.** My paper is closest in spirit to Jacobson, Matthes and Walker (2026). Like them, I hold the surprise and the method fixed and change one measurement. They change how often the outcome is measured. I change how the channel from policy to the economy is measured, and my main result is for activity. Bauer and Swanson (2023*b*) and Jarociński and Karadi (2020) improve the surprise itself. In my design a surprise enters only through its immediate effect on the two-year yield and the FCI and through its timing, so their fixes and mine can be combined.

**Financial channels and risk appetite.** Gertler and Karadi (2015) and Caldara and Herbst (2019) emphasize financial channels too. I use a broad index measured relative to $r^*$ and leave credit spreads out of the baseline. Studies of the "non-yield" shock treat policy-driven moves in stock prices and the dollar as a second shock (Bauer, Schmeling and Schrimpf, 2026; Boehm and Kroner, 2024). Boehm, Kim and Kroner (2026) select the combination of two yield-curve surprises and the non-yield shock that best accounts for macroeconomic fluctuations. They choose the direction of the shock for its macroeconomic content. I fix the conventional direction, news about the path of interest rates, and ask which financial variables are needed to trace it. Caballero, Caravello and Simsek (2026) propose a neutral level of the FCI itself; gaps to that level are a natural extension.

**The power of policy and the Phillips curve.** Policy shocks became less powerful after the early 1980s. The literature attributes this to a more systematic policy response and better-anchored expectations (Boivin and Giannoni, 2006; Boivin, Kiley and Mishkin, 2010; Primiceri, 2005), or to a failure of standard identification once policy became forward-looking (Barakchian and Crowe, 2013). Recent work finds a small Phillips-curve slope (Hazell et al., 2022; Del Negro et al., 2020; Ball and Mazumder, 2019; Fitzgerald et al., 2024), hard to identify when policy offsets demand shocks (McLeay and Tenreyro, 2020) and understated by conventional methods (Barnichon and Mesters, 2020). Barnichon and Mesters (2021) measure the tradeoff by the ratio of cumulative inflation to cumulative unemployment after policy shocks and find it much flatter after 1990 than before. I add evidence of this kind and show that the measured tradeoff depends on how the stance of policy is measured.

## 2 Empirical Approach

The model has three pieces. A surprise moves the two-year yield and the FCI by amounts measured in the window: the vector $b$ in equation (2). A daily VAR, equation (3), carries those two series forward. Their monthly averages then enter a monthly VAR for unemployment and inflation, equation (4). Appendix I gives the construction of the series, the algorithm for tracing responses and the bootstrap, and Appendix J lists the data.

## 2.1 Measuring Financial Conditions

The FCI is the market index of Companion note (2026): a weighted sum of the 10-year Treasury yield, stock valuations and the real broad dollar. Indexes of this kind differ mainly in where the weights come from (Hatzius et al., 2010). FCI-G takes them from large macroeconomic models (Ajello et al., 2023). Mine come from event-study regressions, long used to measure how asset prices respond to policy (Kuttner, 2001; Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018).

Policy news is the first principal component of the changes in federal funds and eurodollar futures around the announcement, rescaled so that one unit moves the two-year yield 1 percentage point (Nakamura and Steinsson, 2018). I regress the window changes in the 10-year yield, the S&P 500 and the dollar on it and weight the three by how precisely each recovers the policy component. Each of the three accounts for about a third of the index's response to policy (Table 9). Because the weights are scaled to the policy news, a policy move that raises the two-year yield 1 percentage point raises the index by 1 unit.

The index comes in three versions. The nominal version uses the nominal 10-year yield and the log of the cyclically adjusted price-earnings ratio (CAPE). The real version subtracts 10-year expected inflation from the yield. The version relative to $r^*$ also subtracts $r^*$ from the yield, and it compares the earnings yield, the inverse of CAPE, with $r^*$: stocks are expensive, and conditions loose, when the earnings yield is low relative to the neutral rate. That comparison enters with its sign reversed and is rescaled to the units of log CAPE, so the stock weight keeps its meaning (Appendix A). The dollar is the real broad dollar in all three versions, and the two-year yield is adjusted in the same three ways.

For $r^*$ I use the New York Fed's one-sided Laubach–Williams estimate (Holston, Laubach and Williams, 2023). It is a filtered estimate that uses data only up to each quarter, from the August 2026 vintage with parameters estimated on the full sample, so it is not a real-time series. Expected inflation and $r^*$ move slowly. The daily series add their interpolated monthly values to daily market prices, so day-to-day moves come from markets.

Figure 2 plots the three versions on a common scale. They move together at business-cycle and higher frequencies; the correlation of 12-month changes is at least 0.90. The real version differs little from the nominal one. The version relative to $r^*$ differs more, and the difference is mostly slow: it is negative in the late 1990s and early 2000s and positive after 2010, as $r^*$ fell.

## 2.2 Why Financial Conditions and $r^*$

Policy reaches spending and prices through many interest rates, asset prices and the dollar. Few households and businesses borrow at the policy rate. They face rates on consumer credit, mortgages and business debt, which arbitrage links to current and expected short rates. Higher rates raise the cost of capital and lower the value of existing capital (Jorgenson, 1963; Hall and Jorgenson, 1967; Tobin, 1969). Lower stock prices lower wealth and consumption (Ando and Modigliani, 1963; Friedman, 1957). A stronger dollar shifts spending toward imports (Fleming, 1962; Mundell, 1963).

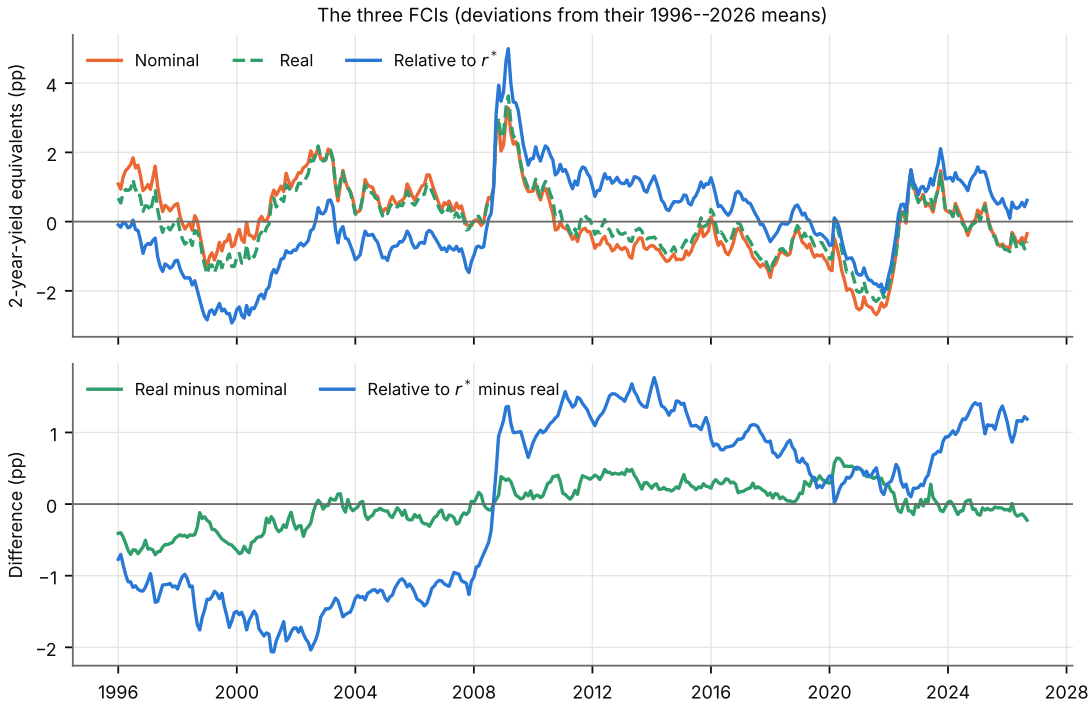



Figure 2: The Three Versions of the FCI

*Notes:* Monthly, in two-year-yield equivalents (a policy move that raises the two-year yield 1 percentage point raises each by one unit in the window). Each series is shown as a deviation from its mean over January 1996 to September 2026. Lower panel: real minus nominal, and relative to $r^*$ minus real.

Modern models of the business cycle contain the same links (Christiano, Eichenbaum and Evans, 2005; Smets and Wouters, 2007; Galí and Monacelli, 2005; Kaplan, Moll and Violante, 2018; Auclert, 2019), and the effects come with a lag (Friedman, 1961).

**What standard VARs leave out.** The workhorse evidence comes from VARs with a few macroeconomic variables and one interest rate (Bernanke and Blinder, 1992; Christiano, Eichenbaum and Evans, 1999). Some studies add a credit spread (Gertler and Karadi, 2015; Caldara and Herbst, 2019), and factor models bring in many more series (Bernanke, Boivin and Eliasz, 2005). Long yields, equity valuations and the dollar are usually absent. These are the variables that carry the effects of policy. Missing variables are an old concern: Sims (1992) argued that the price puzzle arises because the Federal Reserve responds to information that small VARs do not contain (see also Castelnuovo and Surico, 2010). My concern is different, because here the missing variables are the channel itself. If unemployment responds to financial conditions, a VAR in unemployment and the policy rate lets the rate stand in for them. Its error contains past financial shocks, which a VAR with few lags cannot capture (Appendix B; Fernández-Villaverde et al., 2007).

The omission also changes the experiment. A policy surprise that raises the two-year yield 1 percentage point raises the FCI by about 0.99 unit in the window. An ordinary rise of 1 percentage point in the yield, on a day without an announcement, comes with a rise of only 0.28 unit (90 percent interval [0.12, 0.39]; Appendix I). The window loading is close to one because the weights are scaled to policy news. The ordinary loading is not scaled. It is an empirical fact, and it is much smaller. A VAR without financial conditions carries through the ordinary move in the yield. My design puts financial conditions in the system and measures the immediate effect of the surprise on the yield and on financial conditions directly in the window, where policy news is the main force at work.

**Why relative to $r^*$.** What matters in these models is the real rate relative to the neutral rate, not the nominal rate. The same holds for the other parts of financial conditions: a given yield or stock valuation is tighter when $r^*$ is low. The distinction matters most for inflation, which responds to the slack that builds up over time and so to how long conditions stay tight relative to $r^*$. A measure that ignores a falling $r^*$ overstates how tight policy has been, and by more the longer the period.

**Financial conditions and real activity.** Asset prices help forecast output and inflation, but which ones help changes over time (Stock and Watson, 2003). Credit spreads, and especially their excess bond premium component, predict activity well (Gilchrist and Zakrajšek, 2012). Broad indexes summarize many such series (Hatzius et al., 2010; Brave and Butters, 2011), and financial conditions are linked to the risks around growth (Adrian, Boyarchenko and Giannone, 2019), possibly nonlinearly (Kiley, 2020*a*). Federal Reserve staff publish a measure of the effect of financial conditions on growth (Ajello et al., 2023). A correlation between financial conditions and later activity need not be causal, because financial conditions also respond to the economy. The two literatures have complementary strengths. Work on high-frequency surprises identifies policy shocks cleanly but carries them through few variables. Work on financial conditions studies many variables

Table 1: Timing Assumptions

| Frequency | What can move | Assumption | Equation |
|---|---|---|---|
| Window (minutes) | Two-year yield and FCI, by the amounts measured in the window | News about the path of interest rates is the main systematic force in the window, and the only one correlated with the surprise. Unemployment and inflation do not move. | (2) |
| Daily | Two-year yield and FCI, each day after the announcement | They depend on their own recent history and on the unemployment and inflation of this month and last month. | (3) |
| Monthly | Unemployment and inflation, from the month after the announcement | Within a month, unemployment and inflation are set first. They respond to financial variables only with a lag of at least one month. | (4) |

but does not isolate policy shocks. I bring them together.

**Policy news beyond interest rates.** Announcements move stock prices and the dollar by more than interest-rate surprises explain (Boehm and Kroner, 2024; Bauer, Schmeling and Schrimpf, 2026). My windows show the same pattern. The policy-news surprise explains 80 percent of the variance of the move in the two-year yield, but only 32 percent for the 10-year yield, 31 percent for the S&P 500, 34 percent for the dollar and 48 percent for the FCI. The unexplained part is either news unrelated to policy, in which case my estimates capture the effects of policy, or policy news that does not move interest rates, in which case they capture one dimension of policy news and the effects of all of it could be larger. Both readings agree on the channel: policy news reaches the economy through broad financial conditions.

### 2.3 Timing Assumptions and Identification

The shock is the Nakamura–Steinsson policy-news surprise, $s_a$ in equation (6), from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). I use the 252 FOMC announcements from January 1996 to July 2026 with all the window measures available (Appendix I). The window runs from 10 minutes before the statement to 60 minutes after the press conference starts, or 30 minutes around the statement when there is no press conference. Identification comes from timing, at three frequencies. Table 1 summarizes the assumptions.

**The window.** The window move in the FCI is computed with the weights in equation (8):

$$\Delta F_a = w_B \, \Delta y_a^{10} + w_Q \, \Delta \mathrm{SP}_a + w_D \, \Delta \mathrm{DXY}_a. \tag{1}$$

These are the measures available within the day, and the ones used to estimate the weights. Percent changes in the S&P 500 equal percent changes in CAPE within a window, because earnings are fixed. I express all effects per percentage point on the two-year yield in the window. The effect on the FCI is the ratio of two regression slopes on the surprise, an instrumental-variables estimate with

$s_a$ as the instrument:

$$b = \begin{pmatrix} 1 \\ b_F \end{pmatrix}, \qquad b_F = \frac{\sum_a (\Delta F_a - \overline{\Delta F})\,(s_a - \bar{s})}{\sum_a (\Delta y_a^2 - \overline{\Delta y^2})\,(s_a - \bar{s})} = 0.99. \tag{2}$$

This is close to one because the weights are normalized that way. It is not exactly one because this regression includes a constant, which the scaling in equation (6) does not, and because it uses one fewer announcement. Expected inflation and $r^*$ move slowly, so I take the window move to be the same in every version.

**The day.** On the announcement day, the two-year yield and the FCI shift by the vector $b$. I use the window move and not the change over the whole day, because the full-day change contains other news. On later days the two series move as equation (3) implies. They affect each other only through the daily lags, so no ordering between them is needed within the day.

**The month.** Within each month, unemployment and inflation are set first, and markets then respond to them. In the month of the announcement, unemployment and inflation therefore have not moved. They respond from the next month on, through the monthly average of the daily financial series. That average depends on when in the month the announcement falls, since an announcement late in a month affects few days of that month's average. This is how the model takes the timing of announcements into account, as Jacobson, Matthes and Walker (2026) recommend. The ordering is the familiar recursive one, in which slow-moving macroeconomic variables come before financial variables (Bernanke and Blinder, 1992; Christiano, Eichenbaum and Evans, 1999). The daily equations include the current month's unemployment and inflation, even though those figures are published later. I treat the monthly values as a summary of conditions that markets observe as the month unfolds.

**Three steps.** The design separates three steps. The first identifies announcement news: the surprise $s_a$ is policy news measured in a window short enough that news about the path of interest rates is the main systematic force correlated with it. The second measures the immediate financial effects of that news, the vector $b$. The third propagates those effects. After the window, the yield and the FCI evolve as the daily equations estimated on ordinary days imply, and unemployment and inflation respond as the monthly equations estimated on ordinary months imply. Only the first two steps use the announcements.

The third step carries the main assumption. Once the shock is identified, the economy responds to a change in the financial variables in the same way whatever caused it. A shock identified by a Cholesky ordering or an external instrument is carried through the same estimated dynamics as every other shock, and I do the same. Suppose stock prices fall 5 percent in an announcement window because the surprise raises the rate at which future earnings are discounted. On another day they fall 5 percent because earnings forecasts are cut. The first fall raises the cost of capital and lowers wealth with no news about demand. The second may also signal weaker demand, which would lower activity whatever the policy rate does. The model carries both through one equation. If the discount-rate fall is more or less contractionary than the average fall, the estimated propagation

is wrong for the policy surprise. This problem differs from central-bank information, which concerns what $s_a$ reveals in step one and which I set aside, as in the companion note. Propagation could fail even if announcements revealed nothing about the economy.

Two further assumptions are specific to this paper. First, the surprise reaches unemployment and prices only through the yield and the FCI, and has no direct effect on them in the window. Other channels, such as credit spreads and expectations, are left out. So are moves in stock prices and the dollar that are unrelated to $s_a$. The non-yield shock of Boehm and Kroner (2024) is uncorrelated with policy-news surprises, so such moves do not bias $b_F$. Second, the monthly equations see the FCI as a single index. Its weights show how the three components respond to policy news, not how the economy responds to each. I assume the three affect unemployment and prices in proportion to those weights, and Section 3.6 tests that.

A valid external instrument can recover the total response of unemployment and prices with the transmission channels left unobserved, under the usual conditions of a proxy VAR. A yield-only model is not wrong by construction. My claim is narrower. In these data, with 12 lags, a yield-only model does not reproduce the response to the announcement surprise, because the surprise moves financial conditions by much more than the ordinary movement of the yield does. Comparing the two loadings shows that the experiments differ. It does not show how much the difference matters for unemployment and prices. Section 3.4 does.

### 2.4 A Mixed-Frequency VAR

**Structure.** The model has a daily block and a monthly block. Let $x_\tau = (R_\tau, F_\tau)'$ hold the two-year yield and the FCI on day $\tau$, in the same version. Let $m_t = (U_t, \pi_t)'$ hold the two monthly variables. The unemployment gap $U_t$ is the unemployment rate minus the CBO estimate of its natural rate. The CBO estimate does not respond to the surprise, so the response of the gap equals that of the unemployment rate. Inflation is $\pi_t = 1200\,\Delta \ln P_t$, where $P_t$ is the price index for personal consumption expenditures excluding food and energy (core PCE).

The daily block is a VAR that uses averages of the past day, week, month and quarter as regressors (Corsi, 2009):

$$x_\tau = \alpha + \sum_{k\in\{1,5,22,66\}} \Phi_k\, \bar{x}^{(k)}_{\tau-1} + \Gamma_0\, m_{t(\tau)} + \Gamma_1\, m_{t(\tau)-1} + \xi_\tau, \tag{3}$$

where $\bar{x}^{(k)}_{\tau-1}$ is the average of $x$ over the $k$ days through $\tau - 1$, and $t(\tau)$ is the month of day $\tau$. Averages over about a day, a week, a month and a quarter capture persistence with few coefficients. When the FCI is left out, $x_\tau$ is the two-year yield alone.

The monthly block is

$$m_t = c + \sum_{l=1}^{L} A_l\, m_{t-l} + \sum_{l=1}^{L} G_l\, \bar{x}_{t-l} + \zeta_t, \qquad \bar{x}_t = \frac{1}{n_t}\sum_{\tau\in t} x_\tau, \tag{4}$$

with $n_t$ the number of weekdays in month $t$. The model is a mixed-frequency VAR (Ghysels, 2016). The equation contains no current-month term in $\bar{x}_t$, which is the timing assumption of Table 1.

A monthly VAR with the same variables and a recursive ordering is the special case in which every announcement falls on the first day of its month (Appendix C).

**The time trend.** The baseline monthly equations contain a constant and no time trend. Rates drifted down along with $r^*$ over 1996–2026, inflation was low in the 2010s and rose after 2020, and the natural rate of unemployment may have shifted in ways the CBO estimate misses. Section 4.2 adds a spline trend.

**Estimation and inference.** I estimate each block by ordinary least squares, on the same sample: January 1996, the month of the first announcement with all window measures, to September 2026 (369 months, and the weekdays of those months for the daily block). The lagged daily averages and the monthly lags may use earlier observations. 2020 is kept in both blocks. The baseline has $L = 12$ monthly lags, fixed in advance (Appendix E shows 3 and 6). Responses are computed recursively, as deviations from a path with no surprise, and averaged over the positions of the announcements within their months (Appendix I); the response of the price level after $h$ months is $\sum_{j=0}^{h} \tilde{\pi}_j/12$. Intervals come from a block bootstrap with 500 draws (Künsch, 1989; Jentsch and Lunsford, 2019) that resamples calendar months in blocks of 12 and uses together the monthly residuals, the daily observations and the announcements of the months drawn (Appendix D). A response is significant at the 5 or 10 percent level when its 95 or 90 percent interval excludes zero. I report 90 percent intervals, as recent studies in this area do (Bauer and Swanson, 2023*b*; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016), and 68 percent intervals, which Jarociński and Karadi (2020) report alongside them. The FCI weights and $r^*$ are held fixed.

# 3 Baseline Results

## 3.1 Standard Methods after the Early 1980s

Standard identification schemes have not delivered conventional effects of monetary policy in samples that start after the early 1980s. Ramey (2016) estimates the specification of Christiano, Eichenbaum and Evans (1999) on 1983–2007 and finds that "the results change dramatically and imply that increases in the federal funds rate lower the unemployment rate." A hybrid VAR with the Romer–Romer shock (Romer and Romer, 2004) gives, on the same later sample, contractionary shocks that "appear to be expansionary," as Barakchian and Crowe (2013) found for 1988–2007, and a proxy SVAR with the Romer–Romer shock, estimated on 1969–2007, gives a "pronounced, statistically significant price puzzle" (her Figures 1 and 2). Caldara and Herbst (2019) find, for 1994–2007 with a proxy SVAR that uses high-frequency surprises, that without a credit spread the response of industrial production is weak and uncertain (about $-0.2$ percent, with a credible set that includes positive values), against about $-0.75$ percent when the excess bond premium is added.

I start from two standard methods, estimated on my own data and units so that they compare directly with the model below. Both use the monthly unemployment gap, core PCE inflation and

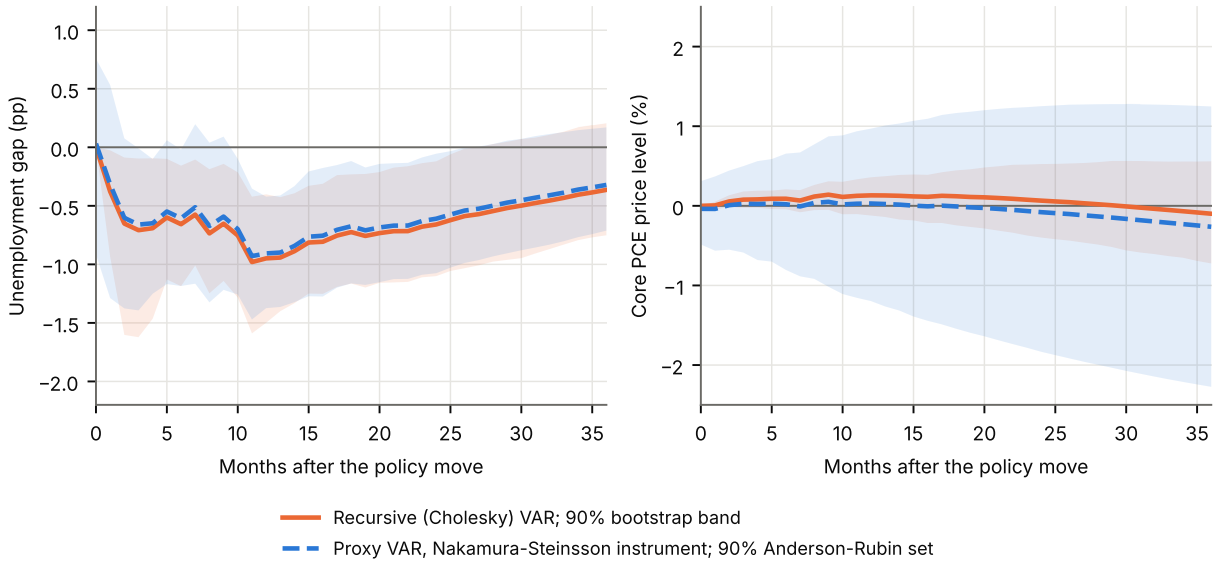



Figure 3: Standard Methods: Responses to a 1 Percentage Point Rise in the Two-Year Yield, 1996–2026

*Notes:* Recursive (Cholesky) VAR and proxy VAR with the Nakamura–Steinsson instrument, in the monthly unemployment gap, core PCE inflation and the two-year yield; 12 lags. Bands: 90 percent (bootstrap for the Cholesky VAR, Anderson–Rubin sets for the proxy VAR, truncated at ±4).

the end-of-month two-year Treasury yield, with 12 lags and no trend. The first is the recursive (Cholesky) VAR of Christiano, Eichenbaum and Evans (1999), with the policy rate ordered last. The second is a proxy VAR (Stock and Watson, 2018; Mertens and Ravn, 2013; Gertler and Karadi, 2015) that uses the Nakamura–Steinsson surprise, summed over the announcements of the month, as an instrument for the policy rate. All responses are to a 1 percentage point rise in the policy rate (Figure 3 and Table 2).

In both, unemployment falls after a tightening, by 0.95 percentage point after 12 months in the Cholesky VAR and 0.91 in the proxy VAR, and it is still lower after 24 months. The price level does not respond. A 1984 start gives the same answer, a funds-rate version on 1984–2007 gives the same sign in smaller numbers, and adding commodity prices, the standard remedy for the price puzzle (Sims, 1992; Ramey, 2016), leaves the response negative (Appendix Table H25). The first-stage $F$ statistic of the proxy VAR is 4.5, so I report exact Anderson–Rubin sets, which stay valid when the instrument is weak (Montiel Olea, Stock and Watson, 2021; Lunsford, 2015; Lewis and Mertens, 2026; Bauer, Schmeling and Schrimpf, 2026). These are the responses when the surprises are carried through the two-year yield alone.

### 3.2 The FCI Model: Baseline Results

Figure 1 shows the baseline. Table 3 compares the responses after 24 months with and without the FCI for the three versions of the yield, on common bootstrap draws. The units are the same in all

Table 2: Standard Methods after the Early 1980s: Responses to a 1 Percentage Point Rise in the Policy Rate

| | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|
| Method | Sample | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Cholesky, 2-year yield | 1996–2026 | 369 | −0.95* | −0.66* | −0.36 | 0.13 | 0.07 | −0.10 |
| | | | [−1.50, −0.40] | [−1.10, −0.12] | [−0.75, 0.21] | [−0.12, 0.36] | [−0.40, 0.52] | [−0.72, 0.56] |
| Cholesky, 2-year yield | 1984–2026 | 513 | −0.61* | −0.36* | −0.16 | 0.11 | 0.15 | 0.13 |
| | | | [−0.99, −0.20] | [−0.59, −0.05] | [−0.37, 0.16] | [−0.06, 0.29] | [−0.16, 0.45] | [−0.30, 0.51] |
| Cholesky, federal funds rate | 1984–2007 | 288 | −0.04 | −0.02 | 0.00 | −0.01 | −0.01 | 0.00 |
| | | | [−0.09, 0.00] | [−0.06, 0.02] | [−0.03, 0.03] | [−0.08, 0.03] | [−0.11, 0.07] | [−0.12, 0.10] |
| Proxy VAR, 2-year yield | 1996–2026 | 369 | −0.91* | −0.61* | −0.32 | 0.03 | −0.08 | −0.26 |
| | | | [−1.37, −0.42] | [−1.08, −0.05] | [−0.71, 0.17] | [−1.20, 0.97] | [−1.83, 1.25] | [−2.27, 1.25] |

*Notes:* Monthly data, 12 lags, constant, no trend; unemployment gap, core PCE inflation and the end-of-month policy rate (the 2-year Treasury yield, or the effective federal funds rate where stated), policy rate ordered last in the Cholesky VARs; price level = cumulated inflation/12. Proxy VAR: Nakamura–Steinsson policy news summed within the month as instrument; first-stage $F = 4.5$. Unit: +1 pp on the policy rate on impact. Brackets: 90% intervals (residual bootstrap for the Cholesky VARs; exact Anderson–Rubin sets for the proxy VAR). * = excludes zero. Specification: baseline/lib/litstyle.py.

Table 3: Responses without and with the FCI after Two Years

| | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|
| Yield measure | Yield alone | Yield and FCI | Difference | Yield alone | Yield and FCI | Difference |
| Nominal | −0.49** | 0.16 | 0.65** | 0.02 | −0.43 | −0.44* |
| | [−0.82, −0.13] | [−0.24, 0.55] | [0.21, 1.08] | [−0.33, 0.42] | [−0.80, 0.02] | [−0.84, −0.04] |
| Real | −0.33* | 0.23 | 0.57** | −0.11 | −0.48** | −0.37** |
| | [−0.64, −0.07] | [−0.11, 0.61] | [0.18, 0.99] | [−0.37, 0.21] | [−0.84, −0.12] | [−0.71, −0.07] |
| Neutral rate ($r^*$-gap) | −0.07 | 0.50* | 0.56** | −0.44** | −0.81** | −0.37** |
| | [−0.42, 0.24] | [0.01, 0.95] | [0.10, 1.06] | [−0.73, −0.05] | [−1.23, −0.33] | [−0.77, −0.07] |

*Notes:* Responses 24 months after a policy move that raises the two-year yield 1 pp in the window, 1996–2026, 12 lags. Differences are computed on common bootstrap draws before rounding. Brackets: 90% intervals; **, *: 5 and 10 percent levels. Responses at 12 and 36 months and at 3 and 6 lags are in Appendix Tables H4 and E1.

results: the end-of-month two-year yield moves 0.92 to 1.01 of the window move, so a 1 percentage point move in the window and a 1 percentage point move in the end-of-month yield in the monthly VARs are the same shock to within about 8 percent.

With the nominal or real yield alone the model gives what the standard methods give. Unemployment falls and the price level does not respond. Measured relative to $r^*$, the yield alone does not raise unemployment but lowers the price level, by 0.44 percent after two years. Adding the FCI changes both responses in every version. Unemployment rises by 0.65, 0.57 and 0.56 percentage point in the nominal, real and neutral-rate versions, and the price level falls by 0.44, 0.37 and 0.37 percent. Measuring both relative to $r^*$ gives the largest responses: unemployment peaks at 0.50 percentage point after 22 months and the price level falls steadily, by 1.11 percent after 36 months.

Appendix Table H4 gives 12, 24 and 36 months, Appendix Table E1 gives 3 and 6 lags, and

Table 4: Monthly VARs with the FCI and the Main Model ($r^*$-gap Two-Year Yield and FCI; 12 Lags)

| | Unemployment gap (pp) | | Price level (%) | |
|---|---|---|---|---|
| | 12 m | 24 m | 12 m | 24 m |
| Proxy VAR | 0.95* | 0.73 | −0.84* | −1.24* |
| | [0.01, 5.87] | [−0.04, 4.55] | [−3.17, −0.15] | [−4.39, −0.24] |
| Window-impact VAR | 0.33 | 0.31 | −0.45* | −0.77* |
| | [−0.25, 0.82] | [−0.25, 0.76] | [−0.71, −0.16] | [−1.26, −0.20] |
| Main model (daily block) | 0.34 | 0.50* | −0.42* | −0.81* |
| | [−0.05, 0.76] | [0.01, 0.95] | [−0.60, −0.20] | [−1.23, −0.33] |

*Notes:* Per +1 pp on the two-year yield. *: the 90% interval excludes zero; intervals in brackets (Anderson–Rubin sets for the proxy VAR; bootstrap for the others). Monthly VARs use end-of-month values.

Appendix Figure H3 shows the paths for the three versions.

### 3.3 Estimating the Impact from Monthly Data

A standard proxy VAR can also carry the FCI. It estimates the immediate effect of the surprise on every variable from monthly residuals and the instrument, instead of measuring it in the window. A window-impact VAR takes the effect on the yield and the FCI from the windows, as the main model does, and sets the effect on unemployment and inflation to zero within the month. It has no daily block, so the timing of announcements within the month is lost. Table 4 gives the results for the yield and the FCI relative to $r^*$.

The proxy VAR has the signs of the main model, with wide sets because the surprise is a weak instrument for monthly residuals (first-stage $F$ of 4.4). The window-impact VAR is close to the main model. Most of what the main model delivers is therefore already in a monthly model that takes the impact from the windows. The daily block adds the timing of announcements and a path for the financial variables at the frequency of the surprise (Section 3.7 and Appendix F). A simulation shows that this is what to expect if the model is the true process: the proxy VAR with the FCI is centred on the true response with a wide range, the proxy VAR without the FCI is biased toward zero, and the window-impact VAR recovers the true response with a narrower range (Appendix Table H29). Measuring the impact in the window gives the same answer with less noise.

### 3.4 Why Adding the FCI Changes the Estimated Response

Adding the FCI changes the experiment and the model at once. A policy surprise raises the FCI by 0.99 unit per percentage point on the yield. A yield-only model has no index to move, and it carries the yield through dynamics estimated without the index. To separate the two changes I compute three paths for the neutral-rate version, on the same sample, calendar, horizon and normalization.

- **Path A** is the baseline: the fitted model with the yield and the FCI, started by the yield rise and the window movement of the FCI.

- **Path B** is the same fitted model, started by the same yield rise and the FCI movement that ordinarily accompanies it, 0.28 unit.
- **Path C** is the matched model with the yield alone.

The ordinary movement is the covariance of the FCI and yield innovations of the daily block divided by the variance of the yield innovation, on days without an announcement. Those are the ordinary observations nearest in timing to the window, but they are close-to-close changes over a whole day. Path B is therefore a model counterfactual, not an observed event.

A − C is the effect of adding the FCI. A − B isolates the initial FCI movement, holding the dynamics fixed. B − C is what remains: the difference associated with including the financial state and re-estimating the propagation of the yield. It has no unique structural interpretation. All three paths use the same bootstrap draws, so A − B and B − C add to A − C at every horizon.

The initial movement accounts for most of the effect. Adding the FCI raises unemployment after 24 months by 0.56 percentage point and lowers the price level by 0.37 percent. The extra initial movement alone accounts for 0.47 and 0.33, and both intervals exclude zero. The remainder, B − C, is 0.10 and 0.04, and neither interval excludes zero. The nominal and real versions give the same split (Table 5). The timing agrees. The A − B contrast for unemployment appears within the first quarter, peaks at about 12 months and fades as the financial movement fades. The contrast for the price level builds steadily, because the price level cumulates inflation. Paths B and C are nearly parallel throughout. The yield-only paths therefore follow path B, and the difference between the yield-only model and the baseline is largely the difference between the ordinary and the announcement movement of financial conditions.

Two points qualify this. The split depends on the ordinary loading. In monthly data it is 0.51, not 0.28, and with that loading A − B is 0.31 and B − C is 0.25 for unemployment; the initial movement remains the larger part. And the initial movement is not the whole story for the size of the baseline response, which also reflects how long the movement lasts. A shorter FCI path would lower the baseline unemployment response after 24 months from 0.50 to 0.17, though the data cannot tell that path from the model's (Appendix F).

### 3.5 The Implied Inflation-Unemployment Tradeoff

What matters for policy is the ratio of the price response to the unemployment response. Following Barnichon and Mesters (2021), I measure it by the Phillips multiplier at horizon $H$, the cumulative price response per unit of cumulative unemployment response:

$$\mathcal{P}_H = \frac{\tilde{P}_H}{\tilde{U}_H}, \qquad \tilde{U}_H = \frac{1}{12}\sum_{h=1}^{H} \tilde{U}_h, \tag{5}$$

where $\tilde{P}_H$ is the response of the core price level after $H$ months, in percent, and $\tilde{U}_H$ is the cumulative response of the unemployment gap, in percentage-point years. Its inverse, $-1/\mathcal{P}_H$, is a sacrifice ratio: the percentage-point years of unemployment needed to lower the price level 1 percent. A more

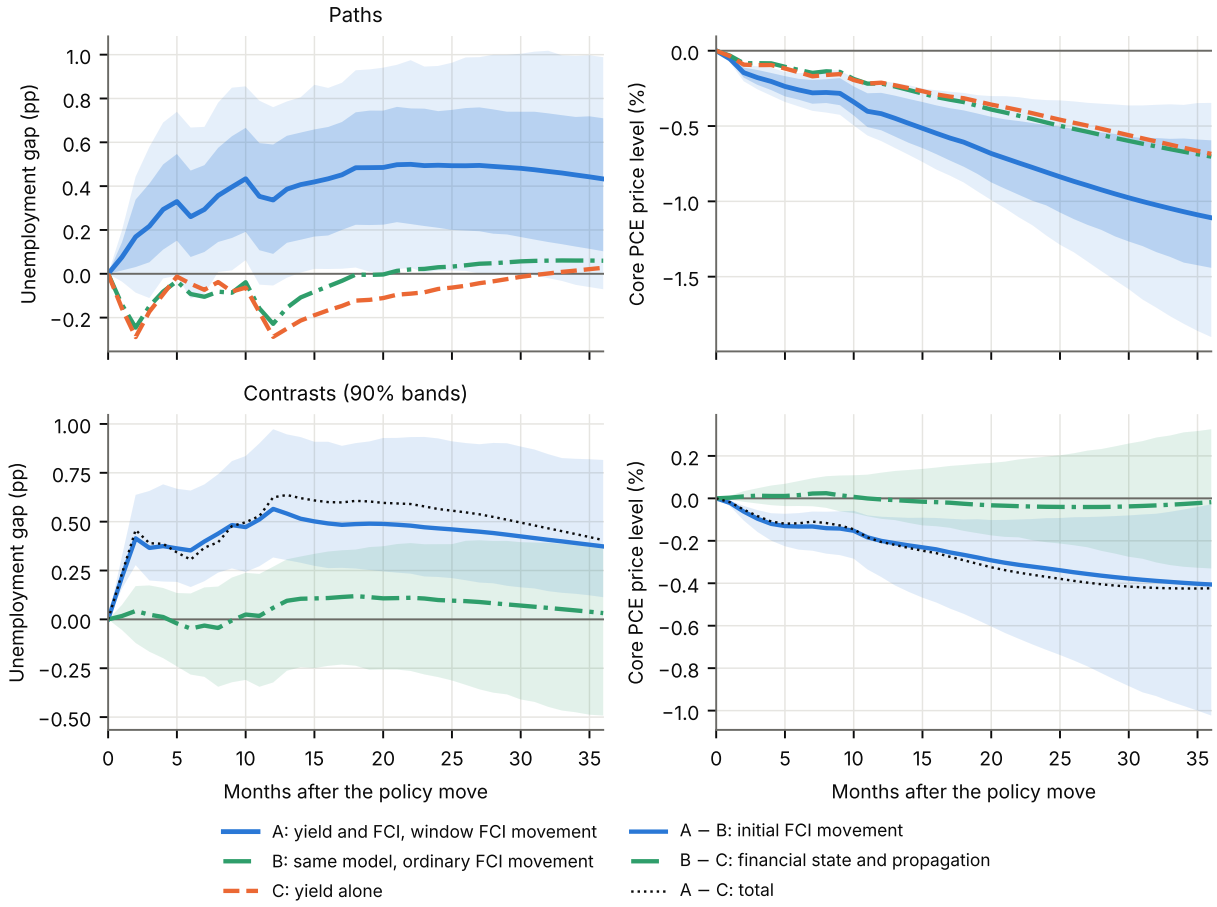



Figure 4: Paths A, B and C and Their Differences

*Notes:* Neutral-rate versions, per +1 pp on the two-year yield in the window; 12 lags. A: yield and FCI, started with the window movement of the FCI. B: same fitted model, started with the FCI movement that ordinarily accompanies the yield. C: yield alone. Bands: 68 and 90 percent intervals (top), 90 percent intervals of paired differences (bottom).

Table 5: The Effect of the FCI: Initial Movement and Propagation

| | Initial FCI movement | Unemployment gap (pp) 24 months | Price level (%) 24 months |
|---|---|---|---|
| *Neutral-rate version (baseline)* | | | |
| A. Yield and FCI, window FCI movement | 0.99 | $0.50^{*}$ | $-0.81^{**}$ |
| | | [0.01, 0.95] | [−1.23, −0.33] |
| B. Same model, ordinary FCI movement | 0.28 | 0.03 | $-0.48^{*}$ |
| | | [−0.40, 0.30] | [−0.75, −0.04] |
| C. Yield alone | – | −0.07 | $-0.44^{**}$ |
| | | [−0.42, 0.24] | [−0.73, −0.05] |
| A − C: effect of adding the FCI | | $0.56^{**}$ | $-0.37^{**}$ |
| | | [0.10, 1.06] | [−0.77, −0.07] |
| A − B: initial FCI movement, dynamics fixed | 0.71 | $0.47^{**}$ | $-0.33^{**}$ |
| | | [0.22, 0.93] | [−0.71, −0.09] |
| B − C: the financial state and propagation | | 0.10 | −0.04 |
| | | [−0.30, 0.38] | [−0.21, 0.20] |
| *Nominal version* | | | |
| A − C: effect of adding the FCI | | $0.65^{**}$ | $-0.44^{*}$ |
| | | [0.21, 1.08] | [−0.84, −0.04] |
| A − B: initial FCI movement, dynamics fixed | 0.73 | $0.44^{**}$ | $-0.30^{**}$ |
| | | [0.17, 0.83] | [−0.63, −0.08] |
| B − C: the financial state and propagation | | 0.21 | −0.15 |
| | | [−0.10, 0.44] | [−0.35, 0.12] |
| *Real version* | | | |
| A − C: effect of adding the FCI | | $0.57^{**}$ | $-0.37^{**}$ |
| | | [0.18, 0.99] | [−0.71, −0.07] |
| A − B: initial FCI movement, dynamics fixed | 0.72 | $0.45^{**}$ | $-0.29^{**}$ |
| | | [0.19, 0.83] | [−0.62, −0.07] |
| B − C: the financial state and propagation | | 0.12 | −0.08 |
| | | [−0.17, 0.35] | [−0.26, 0.11] |

*Notes:* Per +1 pp on the two-year yield in the window. Path A is the baseline model with the yield and the FCI, started with the window movement of the FCI (first column). Path B is the same fitted model started with the FCI movement that ordinarily accompanies a 1 pp innovation in the yield (Appendix I). Path C is the model with the yield alone. All three use the same sample, announcements, calendar and bootstrap draws, so the contrasts are paired and A − B and B − C add to A − C in every draw. Brackets: 90% intervals; $^{**}$, $^{*}$: 5 and 10 percent levels.

Table 6: Implied Trade-Off: Phillips Multiplier $\kappa_H$ (12 Lags)

| Case | FCI | $H = 24$ months | $H = 36$ months |
|---|---|---|---|
| Nominal | none | −0.01 [−0.31, 0.47] | 0.08 [−0.25, 0.62] |
| | nominal | unbounded | unbounded |
| Real | none | 0.14 [−0.23, 0.71] | 0.21 [−0.21, 1.06] |
| | real | unbounded | unbounded |
| r*-gap | none | unbounded | unbounded |
| | r*-gap | −1.07 [−3.78, −0.51] | −0.90 [−3.62, −0.39] |
| Start 1973 | r*-gap | unbounded | unbounded |
| 2020 dropped | r*-gap | −0.95 [−2.00, −0.33] | −0.82 [−2.13, −0.17] |
| 2020 dropped, start 1973 | r*-gap | −1.49 [−2.91, −0.72] | −1.64 [−5.42, −0.72] |
| Ends 2019 | r*-gap | unbounded | unbounded |
| Ends 2019, start 1973 | r*-gap | −1.21 [−6.44, −0.39] | unbounded |
| Kiley r* | r*-gap (Kiley) | −0.22 [−0.71, 0.46] | −0.03 [−0.58, 3.56] |
| Kiley r*, start 1973 | r*-gap (Kiley) | unbounded | unbounded |
| Real, spline trend | real | unbounded | unbounded |
| Real, spline trend, start 1973 | real | unbounded | unbounded |
| r*-gap minus nominal, with FCI | | [−1.12, 6.57] | [−1.08, 8.74] |

*Notes:* $\kappa_H$ = core PCE price level at month $H$ (percent) per percentage-point year of cumulative unemployment-gap response (Barnichon and Mesters 2021); the sacrifice ratio is $-1/\kappa_H$. Point estimate and 90% confidence set (Anderson–Rubin type) in brackets, computed exactly (a wide grid refined by bisection), so a finite end is never an artifact of the grid. An infinite end means that the 90% interval of the cumulative unemployment response includes zero: the data then cannot rule out arbitrarily steep trade-offs on that side; "unbounded": both sides. Last row: paired difference over common announcement draws.

negative $\mathcal{P}_H$ means a steeper Phillips curve. When $\tilde{U}_H$ is close to zero the ratio is not well defined, so I form 90 percent sets for $\mathcal{P}_H$ in the spirit of Anderson and Rubin (1949): the set contains every value $\mathcal{P}$ for which the 90 percent bootstrap interval of $\tilde{P}_H - \mathcal{P}\,\tilde{U}_H$ contains zero. The sets can be unbounded, which shows when the data cannot pin down the tradeoff.

Table 6 gives the results. The tradeoff is measured in the baseline. With the yield and the FCI relative to $r^*$, $\mathcal{P}_{24}$ is −1.07 with a set of [−3.78, −0.51], and $\mathcal{P}_{36}$ is −0.90. The sacrifice ratio after two years is 0.9 percentage-point years of unemployment per 1 percent lower price level, with a set of 0.3 to 2.0. This is not the cost of a permanent one-point decline in annual inflation, and it differs from the three-year ratio of 1.1. In the other versions the data pin the tradeoff down less well. Where adding the FCI leaves the cumulative unemployment response close to zero, the sets are unbounded. A yield-only model that does not raise unemployment says nothing about the cost of lowering inflation.

Barnichon and Mesters (2021) report a multiplier of about −0.15 after three to four years for 1990–2008, estimated with high-frequency surprises, and one that diverges before 1990 and is about −3 after five years, estimated with Romer–Romer shocks. The baseline, at −0.90 after three years, lies between the two.

Table 7: Price and Unemployment Responses with the Components Entered Separately

| | | Monthly equations with | | | | Difference from the index | |
|---|---|---|---|---|---|---|---|
| Start | Response | Index (baseline) | Index, three-component daily block | Equities free | All three free | Equities free | All three free |
| 1996 | Price level, 24 m (%) | −0.81** | −0.72** | −0.52** | 0.06 | 0.20 | 0.78** |
| | | [−1.23, −0.33] | [−1.11, −0.31] | [−0.89, −0.09] | [−0.25, 0.43] | [−0.03, 0.47] | [0.44, 1.12] |
| 1996 | Unemployment gap, 24 m (pp) | 0.50* | 0.40* | 0.33 | 0.22 | −0.08 | −0.19 |
| | | [0.01, 0.95] | [0.03, 0.86] | [−0.06, 0.71] | [−0.26, 0.67] | [−0.37, 0.13] | [−0.63, 0.11] |
| 1973 | Price level, 24 m (%) | −0.63** | −0.65** | −0.60 | −0.60 | 0.05 | 0.05 |
| | | [−0.89, −0.22] | [−0.97, −0.17] | [−0.98, 0.02] | [−0.96, 0.06] | [−0.22, 0.42] | [−0.24, 0.57] |
| 1973 | Unemployment gap, 24 m (pp) | 0.14 | 0.26 | 0.15 | 0.16 | −0.11 | −0.10 |
| | | [−0.17, 0.36] | [−0.09, 0.52] | [−0.31, 0.43] | [−0.27, 0.50] | [−0.50, 0.16] | [−0.49, 0.19] |

*Notes:* $r^*$-gap yield and FCI, 12 lags, per +1 pp on the two-year yield in the window. The last three columns share one daily block (the three components and the two-year yield), so they differ only in the monthly equations: the FCI built from the components as one variable, equities separate with the 10-year yield and the dollar as one variable, or all three separate. The difference columns are computed on common bootstrap draws. The first column is the baseline model with its own daily block. Brackets: 90% intervals; **, *: 5 and 10 percent levels. Nominal and real versions: Appendix Table H23.

### 3.6 Which Financial Variables Carry the Response

The baseline assumes that the 10-year yield, equity valuation and the dollar affect unemployment and prices in proportion to the weights with which they respond to policy news. That is an assumption about the economy, and it can be relaxed and tested. I compare three specifications of the monthly equations. In the first the FCI enters as one variable. In the second equities enter separately and the 10-year yield and the dollar enter as one variable. In the third all three enter separately. The daily block is the same in all three, so the specifications differ only in the monthly equations and share their bootstrap draws (Appendix I).

The unemployment response does not depend on the specification. The price response does, in the sample from 1996. Freeing equities leaves most of it in place (−0.52 percent after 24 months, against −0.72 with the index as one variable). Freeing the dollar as well removes it (0.06). The coefficients show why. In the inflation equation a tighter 10-year yield lowers inflation, as the baseline assumes, and a tighter dollar, holding the yield and equities fixed, has a conditional effect of the opposite sign (−0.66 for the yield and 0.23 for the dollar). A conditional coefficient of this kind does not identify the effect of an exogenous change in the dollar, because the dollar moves with the yield and with equities. It does show that the price result needs the two to enter in proportion.

The tests agree (Table 8). Equality of the yield and dollar coefficients, with equities free, is rejected in the inflation equation from 1996 and not in the unemployment equation. In the sample from 1973 the three specifications give the same price response (−0.65, −0.60 and −0.60) and no test rejects. Equities do not need to be tied to the yield and the dollar. The restriction that matters for prices is the one that ties the dollar to the 10-year yield.

### 3.7 Financial Conditions in the Window

The model assumes that policy reaches unemployment and prices through the yield and the FCI (Table 9). In the window the FCI moves one for one with the yield. Figure 5 shows the responses of

Table 8: Tests of the Proportional-Weights Restriction

| | Three components equal | | Yield and dollar equal, equities free | | Equities equal to yield and dollar | |
|---|---|---|---|---|---|---|
| Version | Unemployment | Inflation | Unemployment | Inflation | Unemployment | Inflation |
| Nominal | 0.984 | 0.053 | 0.821 | 0.024 | 0.978 | 0.142 |
| Real | 0.996 | 0.041 | 0.966 | 0.073 | 0.960 | 0.143 |
| r*-gap | 0.399 | 0.013 | 0.425 | 0.057 | 0.216 | 0.055 |
| Nominal, start 1973 | 0.930 | 0.539 | 0.671 | 0.242 | 0.667 | 0.740 |
| Real, start 1973 | 0.886 | 0.232 | 0.546 | 0.212 | 0.718 | 0.404 |
| r*-gap, start 1973 | 0.408 | 0.374 | 0.134 | 0.561 | 0.358 | 0.371 |

*Notes:* Bootstrap $p$-values of Wald statistics (heteroskedasticity-robust) for the restriction that the coefficients on the weighted components are equal at each lag of the monthly equation, 12 lags: the three components (24 restrictions); the 10-year yield and the dollar with equities free (12); equities and the yield-dollar variable, the yield and dollar being equal (12). Residuals are resampled in moving blocks of 12 months under the restricted equation, 999 draws. Wald statistics and 3 and 6 lags are in Appendix Table H24.

Table 9: Market-Index Weights from Policy News

| Component | Response to policy news | Weight | Share of a policy move |
|---|---|---|---|
| 10-year Treasury yield (pp) | +0.53 (0.06) | +0.609 | 0.33 |
| Equity valuation, 100 ln CAPE (%) | −7.78 (0.93) | −0.051 | 0.40 |
| Real broad dollar, 100 ln (%) | +4.22 (0.53) | +0.065 | 0.28 |

*Notes:* Response of each 30-minute announcement-window change to the policy-news factor (scaled so the 2-year yield moves 1 pp), with robust standard errors; weights from the one-factor GLS of Note A; share = weight × response (the three shares sum to 1).

the two-year yield and the FCI in the baseline. Direct estimates from the days after announcements agree with the model's paths for the yield at every horizon, and for the FCI at 5 and 22 trading days. They have little power at 66 days. The first-month average of the FCI is higher in the data than in the model on the model's clock, and the difference does not change the responses (Appendix F).

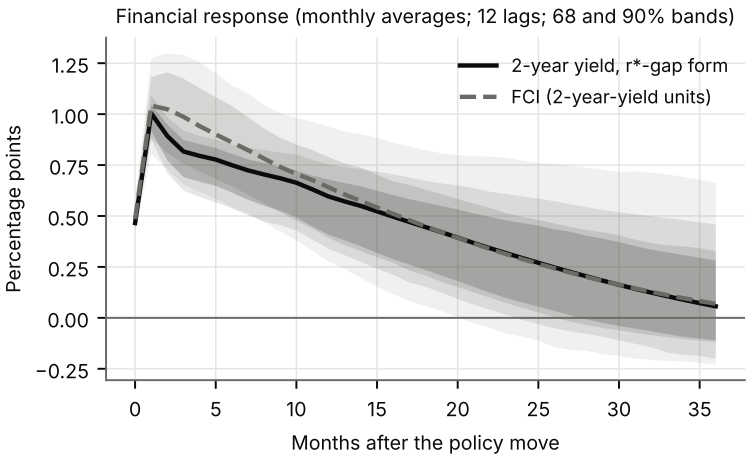



Figure 5: Response of the Two-Year Yield and the FCI to a Policy Move

*Notes:* The FCI is measured relative to $r^*$ and in two-year-yield units; the policy move raises the two-year yield 1 percentage point in the window. Monthly averages; 12 lags; 68 and 90 percent bands.

# 4 Robustness

A result about the transmission of monetary policy can rest on one sample, one estimate of $r^*$, one trend, one set of omitted variables or one surprise series. I vary each in turn, changing one feature of the baseline and keeping the rest (Table 10).

The main results hold across the variations. With the FCI, the wrong-signed unemployment response disappears everywhere: unemployment rises in 12 of the 13 variations that keep the index relative to $r^*$ and never falls significantly, while without the FCI it falls in 11. The price level falls in all 13, and significantly in most. The subsections say what each variation tests.

## 4.1 The Sample

The sample is short, it contains the pandemic, and it ends with the largest tightening in the sample. I start the monthly equations in 1973 instead of 1996 (earlier months use monthly series in place of the daily block), drop March to December 2020, and end all equations in December 2019. In every case the announcements and their window impacts are those of the baseline.

The restrictions remove different data. Dropping 2020 removes ten months from the monthly and daily equations. Ending in 2019 removes 2020–26 from both, and neither isolates the inflation surge of 2021–22. With 2020 dropped or the start moved to 1973, the results are those of the baseline. From 1996, ending the sample in 2019 weakens the price response, from $-0.81$ to $-0.14$ percent after two years, with an interval that includes zero; from 1973 the response is $-0.50$, close to the baseline. Core inflation varied little before 2020 (a standard deviation of 1.1 percentage points for 1996–2019, against 1.8 for 2020–26). The shorter sample has little information on how inflation responds to financial conditions, and the pattern is consistent with the later years supplying it.

Table 10: The Baseline and the Robustness Variations ($r^*$-gap Two-Year Yield; 12 Lags)

| | Yield alone | Yield and FCI | |
|---|---|---|---|
| | Unemployment, 12 m | Unemployment, 24 m | Price level, 24 m |
| Baseline | −0.29 | 0.50* | −0.81** |
| *Sample* | | | |
| Start 1973 | −0.05 | 0.14 | −0.63** |
| 2020 dropped | −0.10 | 0.45** | −0.69** |
| 2020 dropped, start 1973 | 0.17** | 0.24 | −0.76** |
| Ends 2019 | −0.15 | 0.13 | −0.14 |
| Ends 2019, start 1973 | 0.19** | 0.16 | −0.50** |
| *The $r^*$ adjustment and trends* | | | |
| Kiley $r^*$ | −0.53** | 0.28 | −0.18 |
| Kiley $r^*$, start 1973 | −0.11 | −0.04 | −0.37** |
| Real yield and FCI, spline trend | −0.64** | 0.28 | −0.25 |
| Real yield and FCI, spline trend, start 1973 | −0.10 | 0.18 | −0.39** |
| *Credit conditions* | | | |
| Excess bond premium added | −0.08 | 0.52* | −0.78** |
| Excess bond premium added, start 1973 | −0.03 | 0.03 | −0.57** |
| *Weights of the FCI* | | | |
| Three components in place of the FCI | – | 0.22 | 0.06 |
| Three components, start 1973 | – | 0.16 | −0.60 |
| *The surprise* | | | |
| Bauer–Swanson surprise | −0.29 | 0.54* | −0.84** |
| Jarocinski–Karadi surprise | −0.29 | 0.56* | −0.85** |

*Notes:* Per +1 pp on the two-year yield; unemployment gap in percentage points, price level in percent. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero). The variations are described in Sections 4.1 to 4.5; Appendix Tables give each at 12, 24 and 36 months and at 3, 6 and 12 lags. The three-component rows replace the FCI, so they have no yield-alone entry.

Starting in 1973 leaves unemployment close to zero, so the wrong-signed response that the FCI removes belongs to the sample since 1996.

### 4.2 The $r^*$ Adjustment and Trends

The neutral rate is not observed. One variation replaces the Laubach–Williams estimate with Kiley's, in the yield and the FCI alike. Another uses no $r^*$ at all: it keeps the real yield and the real FCI and adds a natural cubic spline in time to every monthly equation. In both, the FCI removes the fall in unemployment. The price response is smaller from 1996 and as in the baseline from 1973. The choice of $r^*$ matters for the size of the price response, not for the sign pattern of unemployment.

### 4.3 Credit Conditions

Credit is the most important channel the baseline leaves out (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019), and a premium that responds to policy could carry part of what the FCI carries. I add the monthly excess bond premium as a third monthly variable. The results are those of the baseline. Without the FCI the unemployment response moves toward zero, in the direction that adding the FCI moves it.

### 4.4 The Surprise

An announcement can reveal the Fed's view of the economy as well as a change in policy, and part of each surprise can be predicted from public data. Two variations change the surprise. One uses the orthogonalized surprise of Bauer and Swanson (2023*b*), which removes the predictable part and is available through 2019 (201 announcements). The other keeps the Nakamura–Steinsson surprise only when the two-year yield and the S&P 500 move in opposite directions in the window, the sign restriction of Jarociński and Karadi (2020) (157 announcements). With the FCI the responses are somewhat larger than in the baseline.

## 5 Conclusion

Interest-rate surprises around Federal Reserve announcements measure policy shocks well, at least the part of policy news that moves interest rates. The hard part has been tracing their effects. I kept the surprise as it is and changed how its effects are traced: through broad financial conditions rather than one interest rate, and relative to the neutral rate. Traced this way, unemployment and prices respond with the standard signs, where the nominal two-year yield alone and the standard methods give the wrong ones. Most of the difference comes from the size of the financial movement that a policy surprise brings, more than three times the movement that accompanies an ordinary rise in the yield, and little from re-estimating the dynamics.

The most durable finding is about unemployment. Once policy news is traced through financial conditions, a tightening does not lower unemployment at any horizon in the baseline or in any

variation. The policy interpretation is the cost of price restraint: about 0.9 percentage-point years of unemployment for each 1 percent lower core price level after two years. The results support judging the stance of policy by financial conditions relative to $r^*$, as the FOMC did in 2023 (Powell, 2023), and they explain why the same policy rate can be more or less restrictive at different times. The estimates are effects of surprises, not a decomposition of any episode.

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## Appendix A The Financial Conditions Index

**Policy news.** Let $\Delta X_a$ be the $5 \times 1$ vector of changes in the current- and next-meeting federal funds futures and the second to fourth eurodollar futures in the window around announcement $a$. Following Nakamura and Steinsson (2018), let $p_a$ be the first principal component of these changes, signed so that it moves with the fourth eurodollar future. Policy news is $p_a$ rescaled so that one unit moves the two-year yield $y^2$ by 1 percentage point:

$$s_a = \gamma\, p_a, \qquad \gamma = \frac{\sum_a p_a\, \Delta y_a^2}{\sum_a p_a^2}. \tag{6}$$

**Weights.** Let $\Delta A_a$ be the vector of changes in the 10-year yield, the S&P 500 and the dollar index (DXY) in the window. The companion note treats these changes as

$$\Delta A_a = \psi\, s_a + \eta_a, \tag{7}$$

where $\psi$ holds the policy responses and $\eta_a$ is other news, with covariance matrix $\Sigma$. It estimates $\psi$ by regressing each element of $\Delta A_a$ on $s_a$ and a constant, and $\Sigma$ from the residuals, over 253 announcements from January 1996 to September 2026. The weights are those that recover the policy component most precisely (one-factor generalized least squares):

$$w = \frac{\Sigma^{-1}\psi}{\psi'\Sigma^{-1}\psi}. \tag{8}$$

Because $w'\psi = 1$ and $s_a$ is scaled to the two-year yield, a policy move that raises the two-year yield 1 percentage point raises the index by 1 unit. The weights are 0.61 on the 10-year yield, $-0.051$ per percent on stock valuations and 0.065 per percent on the dollar. Each of the three accounts for about a third of the index's response to policy.

**Three versions.** In month $t$, the index in version $v$ is

$$F_t^v = w_B\, B_t^v + w_Q\, Q_t^v + w_D\, D_t, \tag{9}$$

measured as a deviation from its sample mean. Here $D_t$ is 100 times the log of the real broad dollar in every version. With $y_t^{10}$ the 10-year yield, $\pi_t^{10}$ 10-year expected inflation, and $(E/P)_t$ the inverse of the cyclically adjusted price-earnings ratio (CAPE), the other two terms are:

$$\text{Nominal:} \quad B_t = y_t^{10}, \qquad Q_t = 100 \ln \text{CAPE}_t, \tag{10}$$

$$\text{Real:} \quad B_t = y_t^{10} - \pi_t^{10}, \qquad Q_t = 100 \ln \text{CAPE}_t, \tag{11}$$

$$r^*\text{-gap (baseline):} \quad B_t = y_t^{10} - \pi_t^{10} - r_t^*, \qquad Q_t = -\frac{100}{\overline{E/P}}\left[(E/P)_t - r_t^*\right]. \tag{12}$$

In the $r^*$-gap version, stocks are measured by the gap between their earnings yield and $r^*$. The term $100/\overline{E/P}$, with $\overline{E/P}$ the sample average, converts this gap to the units of the log CAPE: since

$d \ln \text{CAPE} = -d(E/P)/(E/P)$, a change in the earnings yield of $x$ points near the mean equals a change of $100x/\overline{E/P}$ in 100 ln CAPE with the opposite sign. The sign reversal and the scaling mean that the weight on stocks is the same in every version. CAPE is a real valuation measure, so stocks do not change between the nominal and real versions. For $r^*$ I use the New York Fed's current Laubach–Williams estimates (Holston, Laubach and Williams, 2023). These "one-sided" estimates use only data up to each date, with the model's parameters estimated on the full sample.

**The two-year yield.** The two-year yield is adjusted in the same three versions. With $y_t^2$ the nominal yield and $\pi_t^2$ two-year expected inflation (from the Cleveland Fed; Haubrich, Pennacchi and Ritchken, 2012),

$$R_t^{\text{nominal}} = y_t^2, \qquad R_t^{\text{real}} = y_t^2 - \pi_t^2, \qquad R_t^{r^*\text{-gap}} = y_t^2 - \pi_t^2 - r_t^*. \tag{13}$$

Expected inflation and $r^*$ are monthly or quarterly and move slowly. The daily series add their interpolated monthly values to daily market prices, so day-to-day moves come from markets (Appendix I).

## Appendix B A Two-Equation Illustration

Let $U_t$ be the unemployment gap, $i_t$ the policy rate and $f_t$ financial conditions, with a higher $f_t$ meaning tighter conditions:

$$U_t = \rho_U \, U_{t-1} + \kappa \, f_{t-1} + \varepsilon_t^U, \tag{14}$$

$$f_t = \rho_f \, f_{t-1} + \lambda \, i_t + \varepsilon_t^f. \tag{15}$$

Unemployment responds to financial conditions and not directly to the policy rate. Financial conditions respond to the policy rate and to other forces, $\varepsilon_t^f$, such as changes in risk appetite. Some of these other forces may themselves be policy news that does not move interest rates (Boehm and Kroner, 2024; Bauer, Schmeling and Schrimpf, 2026). Suppose the economist observes only $U_t$ and $i_t$. To remove $f_t$, lag equation (14) one month, multiply it by $\rho_f$, subtract it from equation (14), and use equation (15). This gives

$$U_t = (\rho_U + \rho_f) \, U_{t-1} - \rho_U \rho_f \, U_{t-2} + \kappa\lambda \, i_{t-1} + \underbrace{\varepsilon_t^U - \rho_f \, \varepsilon_{t-1}^U + \kappa \, \varepsilon_{t-1}^f}_{\text{error}}. \tag{16}$$

The policy rate has no direct effect in this model. It appears in equation (16) because it stands in for financial conditions. The error is a moving average that includes past financial shocks, so a VAR in $U_t$ and $i_t$ with few lags cannot capture it. The VAR is misspecified. With the lags and samples available in practice, this can make the effects of monetary policy harder to measure.

## Appendix C Relation to a Monthly VAR and a Proxy VAR

Suppose the daily block were replaced by a monthly equation for $\bar{x}_t$, and every announcement occurred on the first day of the month. With $Z_t = (m_t', \bar{x}_t')'$, the model would then be a standard monthly VAR with a recursive ordering, the macroeconomic variables first, and impact vector $(0, 0, b')'$. The daily block adds two things: the position of each announcement in its month, and fast dynamics of the financial variables, estimated on many more observations. A proxy VAR (Gertler and Karadi, 2015; Mertens and Ravn, 2013; Stock and Watson, 2018) instead estimates $Z_t = \sum_l C_l Z_{t-l} + \iota_t$ and uses as an instrument $z_t$ the surprises summed over the month. The effect on the FCI per unit effect on the two-year yield is then

$$\frac{\mathrm{Cov}(\iota_{F,t}, z_t)}{\mathrm{Cov}(\iota_{R,t}, z_t)}. \tag{17}$$

The window ratio $b_F$ in equation (2) has the same form, with three differences. It replaces monthly VAR residuals with changes measured inside the window, which do not depend on how strongly the surprise predicts monthly residuals. It sets the entries for unemployment and inflation to zero rather than estimating them. And it dates each shock to its day rather than summing shocks over a month. Section 3.3 estimates both monthly benchmarks. In large samples a VAR of this kind estimates the same effects as local projections (Plagborg-Møller and Wolf, 2021). In the language of mixed-data sampling, the monthly equation gives equal weights to the days within a month and unrestricted weights to the months (Ghysels, Sinko and Valkanov, 2007; Foroni, Marcellino and Schumacher, 2015).

## Appendix D The Bootstrap

Every interval in the note comes from a block bootstrap. Each draw resamples the 369 calendar months from January 1996 in moving blocks of 12 months and uses together the monthly residuals, the daily observations and the announcements of the months drawn, which keeps the dependence between them. The announcements drawn give new values of $b$ and new positions. The monthly equations are re-estimated on a series rebuilt recursively from the drawn residuals, with the macro lags taken from the simulated series and the financial lags from the data, and the daily block is re-estimated on the drawn daily observations. When the monthly equations start in 1973, the residuals before 1996 are drawn in blocks of 12 months of their own. A month without a monthly observation (2020 dropped, or a sample ending in 2019) contributes its daily observations and announcements but no residual. A simpler bootstrap resamples the three sources of sampling variation separately: the announcements and the monthly residuals independently across months (with the monthly lags held at their values in the data), and the daily observations in moving blocks of 66 days. I ran it with 12 lags and 500 draws for the nominal, real and $r^*$-relative two-year yield, each without and with the FCI of the same version. Across the 36 responses at 12, 24 and 36 months, its 90 percent intervals are wider than those of the block bootstrap in 36, by a median of

1.14 times (range 1.03 to 1.34). The two agree on whether a response is significant at the 10 percent level in 34 of the 36 responses and at the 5 percent level in 30. The price response of the baseline is significant at the 5 percent level at every horizon under both. The 90 percent interval of the baseline unemployment response after 24 months is [0.01, 0.95] under the block bootstrap and [0.02, 1.07] under the simpler one.

## Appendix E Lag Length

The baseline uses 12 monthly lags, fixed in advance. Table E1 repeats the main results with 3 and 6 lags. The sample, the announcements and the bootstrap are the same at every lag length: the block bootstrap with 500 draws. The tables that follow give every case at 3, 6 and 12 lags.

The direction of every result is the same at every lag length. With the FCI, the price level falls in all 48 responses (16 cases at 12, 24 and 36 months), and one response of unemployment is significantly negative, at 3 lags only (Kiley's $r^*$ from 1973, after 24 months: $-0.11$ [$-0.28$, $-0.01$]). Unemployment rises in 34 of the 48 responses with 3 lags, 43 with 6 and 42 with 12. Without the FCI, the Cholesky and proxy VARs show a fall in unemployment at every lag length; the Cholesky fall at 12 months is significant at all three. The weights test gives the same verdicts: in the sample from 1996 it rejects proportional weights in the inflation equation at the 10 percent level and does not reject them in the unemployment equation at the 5 percent level, and in the sample from 1973 it rejects in neither. In the simulation, the median proxy-VAR estimate of the unemployment response with the FCI is close to the true response at every lag length.

The size of the responses and the strength of the evidence increase with the lag length. The median price response with the FCI across the 48 responses is $-0.21$ with 3 lags, $-0.35$ with 6 and $-0.44$ with 12. The baseline price response at 36 months is $-0.27$, $-0.87$ and $-1.11$ percent. The number of significant price responses is 4, 29 and 37 of 48. The Phillips multiplier is bounded at 12 lags only; with 3 or 6 lags the data do not rule out arbitrarily steep trade-offs.

Table E1: The Main Results with 3, 6 and 12 Monthly Lags

| | 3 lags | 6 lags | 12 lags |
|---|---|---|---|
| *Standard methods (per +1 pp on the policy rate)* | | | |
| Cholesky, 2-year yield: unemployment gap, 12 m | −0.64* [−1.01, −0.15] | −0.56* [−0.97, −0.12] | −0.95* [−1.50, −0.40] |
| Cholesky, 2-year yield: price level, 36 m | 0.09 [−0.49, 0.47] | −0.08 [−0.74, 0.36] | −0.10 [−0.72, 0.56] |
| Proxy VAR: unemployment gap, 24 m | −0.43* [−0.64, −0.10] | −0.36 [−0.76, 0.34] | −0.61* [−1.08, −0.05] |
| *FCI model, r*-gap 2-year yield (baseline)* | | | |
| With FCI: unemployment gap, 24 m | 0.13 [−0.28, 0.49] | 0.36 [−0.12, 0.71] | 0.50* [0.01, 0.95] |
| With FCI: price level, 24 m | −0.25 [−0.61, 0.16] | −0.63* [−1.01, −0.16] | −0.81* [−1.23, −0.33] |
| With FCI: price level, 36 m | −0.27 [−0.75, 0.26] | −0.87* [−1.38, −0.18] | −1.11* [−1.90, −0.35] |
| Without FCI: unemployment gap, 12 m | −0.17 [−0.44, 0.14] | −0.09 [−0.38, 0.21] | −0.29 [−0.60, 0.11] |
| Phillips multiplier, 24 m | −0.65 [unbounded] | −1.07 [unbounded] | −1.07 [−3.78, −0.51] |
| *All cases with the FCI in Sections 3 and 4 (16 cases, 3 horizons, 48 responses each)* | | | |
| Price level falls | 48 of 48 | 48 of 48 | 48 of 48 |
| Price level falls significantly | 4 of 48 | 29 of 48 | 37 of 48 |
| Unemployment gap rises | 34 of 48 | 43 of 48 | 42 of 48 |
| Unemployment gap rises significantly | 4 of 48 | 7 of 48 | 10 of 48 |
| *Financial weights and simulation (r*-gap)* | | | |
| Proportional weights, 1996: bootstrap $p$, unemployment equation | 0.201 | 0.236 | 0.399 |
| Proportional weights, 1996: bootstrap $p$, inflation equation | 0.017 | 0.050 | 0.013 |
| Proportional weights, 1973: bootstrap $p$, inflation equation | 0.609 | 0.752 | 0.374 |
| Proxy VAR with FCI, simulated: median unemployment, 12 m (true value) | 0.25 (0.25) | 0.36 (0.33) | 0.35 (0.33) |
| Proxy VAR with FCI, simulated: first-stage $F$, median | 12.2 | 11.8 | 11.1 |

*Notes:* Same sample (1996 to 2026), announcements and block bootstrap (500 draws) at every lag length. Responses per +1 pp on the policy rate; **, *: 95% and 90% intervals exclude zero; 90% intervals in brackets. Counts (at the 10 percent level): the 16 cases with the FCI in the baseline and Section 4.1, 2, 3 and 5 (components excluded), responses at 12, 24 and 36 months.

# Appendix F Financial Persistence beyond the Announcement Window

The monthly equations respond to the monthly average of the daily financial variables, so the model's answer depends on how long the window movement of the yield and the FCI lasts. The daily block (equation (3)) is estimated on ordinary days. I check it against the data in the days after announcements.

**Definitions.** Day 0 is the announcement day and day $h$ is $h$ trading days later. The model sets the level of each variable on day 0 equal to its window movement and propagates it from there. I build the direct estimate on the same clock. The change on day $h$ is the window movement plus the change in the closing level from day 0 to day $h$, $D_h = w_a + (x_{a,h} - x_{a,0})$, so day 0 is the window movement itself. The endpoints are $D_5$, $D_{22}$ and $D_{66}$. The average is the mean of $D_0, \ldots, D_{21}$, 22 trading days with the announcement day included, which corresponds to the first month of the monthly average. The model works on weekdays and the data on trading days, so 66 weekdays are about 63 to 65 trading days. As a check I also build the change from the close before the announcement, $x_{a,h} - x_{a,-1}$. The two differ by the part of the announcement day's close-to-close move that the window does not contain, the window-to-close gap in the notes of Table F2.

**Estimation.** Each $D_h$ is regressed on the policy-news surprise over the 251 announcements with 66 trading days after them, the common set for all horizons, and the slope is divided by the slope of the window movement of the two-year yield on the same announcements. The unit is then a one-point rise in the yield in the window, as in the model. Announcements less than 66 trading days apart overlap. Two things address this. Intervals come from moving blocks of 12 calendar months, resampling the announcements of the months drawn and re-estimating the normalization, the scheme of the main bootstrap. And a second estimate adds the sum of the surprises of the announcements that follow within the horizon as a control. The model paths are those of the baseline, evaluated at the same trading-day horizons, with the same bootstrap draws.

**Results.** Table F2 and Figure F1 give the direct estimates and the model. For the two-year yield the direct estimates are 0.82 at 5 days, 0.81 at 22 days and 0.60 at 66 days, against 0.99, 1.01 and 0.81 in the model; every model value lies inside the 90 percent interval of the direct estimate, and so does the model's first-month average (1.02 against 0.89, interval [0.52, 1.16]). For the FCI the direct estimates are 1.31 at 5 days and 1.01 at 22 days against the model's 1.07 and 1.04, again inside the intervals. At 66 days the direct estimate is 0.32 with an interval of $[-0.57, 1.35]$, which contains the model's 0.98 and also a complete reversal: the data cannot distinguish persistence from decay at that horizon. Adding the surprises of later announcements as controls changes none of these conclusions.

The one difference is the first-month average of the FCI. On the model's clock the direct estimate is 1.40 (interval [1.12, 1.79]) and the model's value 1.05 lies below it. The difference comes from the announcement day. The close-to-close move of the FCI on the announcement day is $-0.28$ $[-0.44, -0.15]$ below the window move: part of the window's movement in stock prices, long yields and

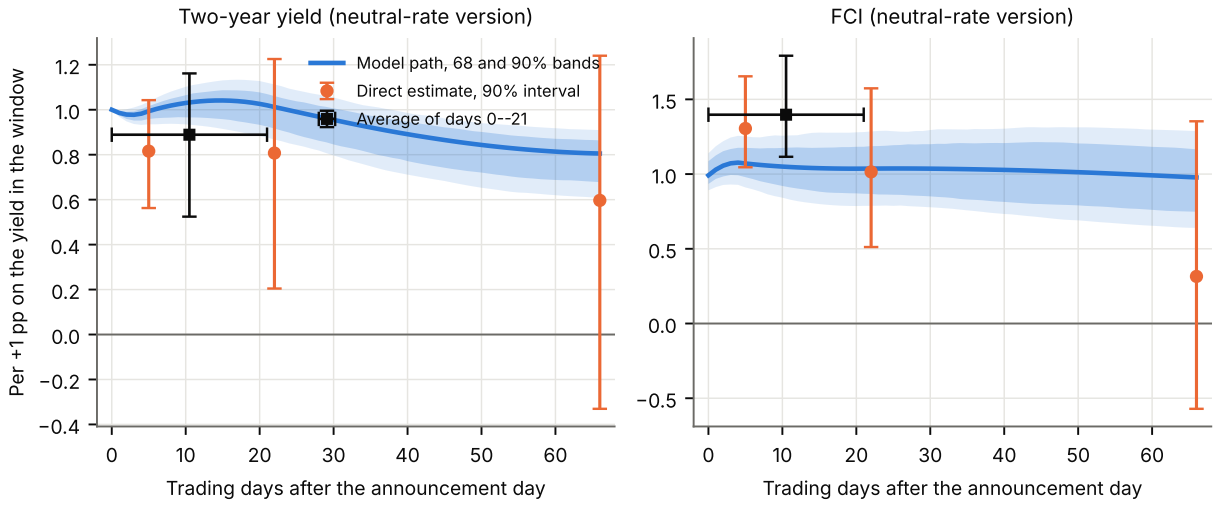



Figure F1: Financial Persistence: Direct Estimates and the Model

*Notes:* Per +1 pp on the two-year yield in the window, $r^*$-gap versions. Blue: the model's daily path after the announcement day (day 0), with 68 and 90 percent bands. Orange: direct estimates at 5, 22 and 66 trading days with 90 percent intervals. Black square: the average of days 0 to 21, with its interval; the horizontal bar shows the days averaged. The calendar-month averages of Figure 5 measure something different.

the dollar has reversed by the close. Built from the close before the announcement, the average is 1.12 and the model lies inside the interval. The window-to-close gap is also a statement about the starting point: if the FCI started the model at the closing level and not at the window level, the 24-month responses would be −0.18 percentage point for unemployment and 0.13 percent for the price level different, which is a third of path A − C in Section 3.4 and does not change its sign.

**What the differences imply for the macro response.** A sensitivity calculation traces the macro response through the monthly equations of the baseline, holding the daily block fixed. If the event month's average of the FCI and the yield are those of the direct estimate on the model's clock (the FCI 0.35 higher and the yield −0.13 lower), the responses after 24 months change by 0.00 percentage point for unemployment and 0.00 percent for the price level. If the same excess lasted through the first three months, the changes are 0.00 and −0.01. The first-month average therefore does not matter for the results. Persistence at longer horizons would. If the FCI path from the second month on were scaled by the ratio of the direct to the model estimate at 66 days (0.32), the responses after 24 months would be 0.17 for unemployment and −0.55 for the price level, against 0.50 and −0.81; with the ratio at the ends of the 90 percent interval (-0.58 and 1.38) they would be −0.26 and −0.20, and 0.68 and −0.95. The data at 66 days cannot narrow that range. The check shows that the model's persistence is consistent with the days after announcements. It cannot show that it is right beyond the first month.

Table F2: Financial Persistence: Direct Estimates and the Model

| Series | | Direct estimate | With later announcements controlled | Model |
|---|---|---|---|---|
| Two-year yield | Day 5 | 0.82 | 0.80 | 0.99 |
| | | [0.56, 1.04] | [0.55, 1.04] | [0.94, 1.04] |
| | Day 22 | 0.81 | 0.76 | 1.01 |
| | | [0.20, 1.23] | [0.18, 1.21] | [0.87, 1.11] |
| | Day 66 | 0.60 | 0.66 | 0.81 |
| | | [−0.33, 1.24] | [−0.22, 1.42] | [0.61, 0.91] |
| | Average, days 0–21 | 0.89 | 0.85 | 1.02 |
| | | [0.52, 1.16] | [0.49, 1.14] | [0.93, 1.08] |
| FCI | Day 5 | 1.31 | 1.33 | 1.07 |
| | | [1.05, 1.65] | [1.07, 1.69] | [0.91, 1.25] |
| | Day 22 | 1.01 | 1.05 | 1.04 |
| | | [0.51, 1.57] | [0.54, 1.66] | [0.79, 1.29] |
| | Day 66 | 0.32 | 0.30 | 0.98 |
| | | [−0.57, 1.35] | [−0.60, 1.43] | [0.64, 1.29] |
| | Average, days 0–21 | 1.40 | 1.48 | 1.05 |
| | | [1.12, 1.79] | [1.22, 1.84] | [0.86, 1.25] |

*Notes:* Response per +1 pp on the two-year yield in the window, $r^*$-gap versions, 251 announcements with 66 trading days after them. Day 0 is the announcement day; the change on day $h$ is the window move plus the change in the close from day 0 to day $h$, the convention of the model, and the average is over days 0 to 21 (22 trading days). Direct estimate: slope of that change on the policy-news surprise divided by the window slope of the yield; second column adds the surprises of announcements within the horizon. 90% intervals from 2,000 draws of moving blocks of 12 calendar months with the normalization re-estimated. Model: the baseline of Table 3 on weekdays; 90% bootstrap interval. Close minus window on the announcement day (the window-to-close gap): −0.12 [−0.32, 0.11] for the yield and −0.28 [−0.44, −0.15] for the FCI, per +1 pp in the window.

## Appendix G A Direct Projection on the Surprise

The main model imposes that the surprise reaches unemployment and prices through the yield and the FCI. A direct projection imposes nothing: all channels operate, and no variable of the financial block enters. The surprise, summed over the announcements of a month, is the regressor, and the unemployment gap, or the cumulated core inflation of the months from the announcement month to month $t+h$, is the outcome. The specification, the controls and the inference are in the notes of Table G3. The existing local-projection code of this project (`experimental/rstar_case/lp_unemp.py`) projects unemployment on past values of an index, not on the surprise, and the earlier distributed-lag regressions of the surprise on the monthly outcome (`experimental/fin_shocks/legacy/direct_mfdl.py`) use event-day shocks with a restricted lag shape. Neither instruments a monthly yield innovation. The projection here is the direct response, scaled by the window slope of the yield, with no first stage.

**Results.** The direct projection has the features of the earlier direct estimates. With macro lags the unemployment response is $-2.86$ percentage point after 12 months and $-3.27$ after 24, and the price level 0.67 and 2.18 percent. The sign of unemployment is the wrong one and the price response is not negative. The estimates are noisy: the 90 percent bootstrap intervals include zero for all four ([$-4.84$, 0.70] for unemployment after 12 months), and Newey–West intervals exclude zero for unemployment at both horizons and for the price level after 24 months. When the lags of the yield and the FCI are added to the controls, the estimates change sign for unemployment (0.75 and 0.26) and the price level stays above zero (0.28 and 0.73); no interval excludes zero. The main model's responses lie inside the bootstrap intervals of both projections at both horizons, except for the price level after 24 months in the first projection.

The direct projection allows all transmission channels to operate, including the exchange rate, expectations and the response of policy to the economy after the announcement, and nothing is imposed. A perverse estimate does not show that omitting the FCI biases the result, and I do not control for the contemporaneous FCI. It shows that the direct response of the economy to the surprise, taken at face value, does not have the signs that the theory gives, and that the data do not pin it down. The sign change when lagged financial conditions are added is a feature of the estimates, not a test of the model. The comparison is informative for identification. The announcement surprise by itself gives an imprecise direct macro estimate; the paper obtains a more informative estimate by imposing a model of financial propagation. The proxy VAR of Table 4 is the benchmark that keeps the monthly dynamics and is therefore the comparison for the main model.

Table G3: A Direct Projection on the Policy Surprise

| Outcome | Months | Direct projection | Direct projection, financial lags | Main model | Proxy VAR |
|---|---|---|---|---|---|
| Unemployment gap (pp) | 12 | −2.86 | 0.75 | 0.34 | 0.95* |
| | | [−4.84, 0.70] | [−1.46, 4.08] | [−0.05, 0.76] | [0.01, 5.87] |
| | 24 | −3.27 | 0.26 | 0.50* | 0.73 |
| | | [−6.32, 0.75] | [−2.64, 3.01] | [0.01, 0.95] | [−0.04, 4.55] |
| Price level (%) | 12 | 0.67 | 0.28 | −0.42** | −0.84* |
| | | [−0.88, 1.46] | [−1.36, 1.08] | [−0.60, −0.20] | [−3.17, −0.15] |
| | 24 | 2.18 | 0.73 | −0.81** | −1.24* |
| | | [−0.52, 4.21] | [−1.64, 3.01] | [−1.23, −0.33] | [−4.39, −0.24] |

*Notes:* Direct projection: the outcome in month $t+h$ (unemployment gap, or the cumulated core PCE inflation of months $t$ to $t+h$ divided by 12, as in the main model) regressed on the sum of the policy-news surprises of announcement month $t$, with a constant and 12 lags of the unemployment gap and inflation (and, in the second column, 12 lags of the monthly averages of the $r^*$-gap yield and FCI); all controls are dated before month $t$ and the contemporaneous FCI is not included. January 1996 to September 2026, nothing dropped. The coefficient is divided by the window slope of the yield on the surprise (1.01), so the unit is +1 pp on the two-year yield in the window and there is no first stage. Brackets: 90% intervals from moving blocks of 12 (12 months) or 24 (24 months) calendar months with the scaling re-estimated. Main model and proxy VAR: Tables 3 and 4. *: the 90% interval excludes zero.

# Appendix H Additional Tables

Table H10: Baseline: 1996 to the Present, All Data in, No Trend

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | none | 3 | 369 | −0.59** | −0.45** | −0.36** | 0.11 | 0.08 | 0.05 |
| | | | | [−0.81, −0.29] | [−0.72, −0.13] | [−0.66, −0.04] | [−0.06, 0.28] | [−0.18, 0.35] | [−0.29, 0.44] |
| | | 6 | 369 | −0.55** | −0.42** | −0.30* | 0.07 | 0.01 | −0.10 |
| | | | | [−0.80, −0.24] | [−0.67, −0.10] | [−0.59, 0.00] | [−0.11, 0.26] | [−0.31, 0.32] | [−0.52, 0.36] |
| | | 12 | 369 | −0.81** | −0.49** | −0.31 | 0.09 | 0.02 | −0.12 |
| | | | | [−1.20, −0.38] | [−0.82, −0.13] | [−0.69, 0.06] | [−0.11, 0.28] | [−0.33, 0.42] | [−0.61, 0.46] |
| | nominal | 3 | 369 | −0.02 | 0.07 | 0.05 | −0.10 | −0.27 | −0.37 |
| | | | | [−0.33, 0.27] | [−0.32, 0.38] | [−0.32, 0.41] | [−0.31, 0.10] | [−0.59, 0.08] | [−0.81, 0.09] |
| | | 6 | 369 | 0.01 | 0.17 | 0.17 | −0.14 | −0.39 | −0.60* |
| | | | | [−0.29, 0.35] | [−0.19, 0.54] | [−0.22, 0.58] | [−0.34, 0.09] | [−0.77, 0.01] | [−1.15, −0.03] |
| | | 12 | 369 | −0.05 | 0.16 | 0.22 | −0.13 | −0.43 | −0.71 |
| | | | | [−0.50, 0.35] | [−0.24, 0.55] | [−0.20, 0.67] | [−0.33, 0.09] | [−0.80, 0.02] | [−1.22, 0.01] |
| Real | none | 3 | 369 | −0.34** | −0.33** | −0.29** | −0.05 | −0.07 | −0.09 |
| | | | | [−0.58, −0.14] | [−0.58, −0.08] | [−0.58, −0.05] | [−0.17, 0.11] | [−0.27, 0.21] | [−0.37, 0.32] |
| | | 6 | 369 | −0.33** | −0.29* | −0.22 | −0.01 | −0.09 | −0.19 |
| | | | | [−0.57, −0.12] | [−0.55, −0.03] | [−0.51, 0.03] | [−0.17, 0.15] | [−0.39, 0.20] | [−0.58, 0.26] |
| | | 12 | 369 | −0.60** | −0.33* | −0.22 | −0.03 | −0.11 | −0.21 |
| | | | | [−0.96, −0.27] | [−0.64, −0.07] | [−0.55, 0.05] | [−0.18, 0.14] | [−0.37, 0.21] | [−0.61, 0.28] |
| | real | 3 | 369 | 0.12 | 0.06 | 0.00 | −0.20* | −0.31 | −0.37 |
| | | | | [−0.17, 0.37] | [−0.26, 0.36] | [−0.30, 0.31] | [−0.37, −0.01] | [−0.59, 0.01] | [−0.73, 0.05] |
| | | 6 | 369 | 0.14 | 0.19 | 0.14 | −0.20* | −0.42** | −0.60** |
| | | | | [−0.16, 0.41] | [−0.13, 0.55] | [−0.19, 0.53] | [−0.40, −0.02] | [−0.80, −0.08] | [−1.10, −0.11] |
| | | 12 | 369 | 0.09 | 0.23 | 0.21 | −0.23** | −0.48** | −0.66** |
| | | | | [−0.36, 0.43] | [−0.11, 0.61] | [−0.14, 0.62] | [−0.41, −0.06] | [−0.84, −0.12] | [−1.24, −0.10] |

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 369 | −0.17 | −0.16 | −0.14 | −0.12 | −0.15 | −0.17 |
| | | | | [−0.44, 0.14] | [−0.45, 0.16] | [−0.42, 0.15] | [−0.27, 0.06] | [−0.43, 0.14] | [−0.54, 0.25] |
| | | 6 | 369 | −0.09 | −0.06 | −0.01 | −0.16 | −0.35* | −0.53* |
| | | | | [−0.38, 0.21] | [−0.43, 0.25] | [−0.37, 0.26] | [−0.33, 0.02] | [−0.68, 0.00] | [−1.02, −0.04] |
| | | 12 | 369 | −0.29 | −0.07 | 0.03 | −0.21** | −0.44** | −0.69** |
| | | | | [−0.60, 0.11] | [−0.42, 0.24] | [−0.35, 0.36] | [−0.35, −0.05] | [−0.73, −0.05] | [−1.13, −0.10] |
| | r*-gap | 3 | 369 | 0.26 | 0.13 | 0.01 | −0.20 | −0.25 | −0.27 |
| | | | | [−0.09, 0.63] | [−0.28, 0.49] | [−0.42, 0.40] | [−0.41, 0.04] | [−0.61, 0.16] | [−0.75, 0.26] |
| | | 6 | 369 | 0.34 | 0.36 | 0.26 | −0.33** | −0.63** | −0.87** |
| | | | | [−0.03, 0.65] | [−0.12, 0.71] | [−0.22, 0.65] | [−0.51, −0.11] | [−1.01, −0.16] | [−1.38, −0.18] |
| | | 12 | 369 | 0.34 | 0.50* | 0.43 | −0.42** | −0.81** | −1.11** |
| | | | | [−0.05, 0.76] | [0.01, 0.95] | [−0.07, 0.99] | [−0.60, −0.20] | [−1.23, −0.33] | [−1.90, −0.35] |

*Notes:* Nominal, real and r*-gap 2-year yield; the FCI is the matching index of Note A in each case. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H11: Section 4.1 (a): Monthly Equations Start in 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 642 | 0.01 | 0.05 | 0.05 | −0.09 | −0.15 | −0.20 |
| | | | | [−0.10, 0.15] | [−0.09, 0.21] | [−0.08, 0.21] | [−0.13, 0.18] | [−0.17, 0.46] | [−0.20, 0.77] |
| | | 6 | 639 | 0.00 | 0.03 | 0.03 | −0.12 | −0.31 | −0.53 |
| | | | | [−0.11, 0.12] | [−0.11, 0.19] | [−0.12, 0.16] | [−0.22, 0.04] | [−0.49, 0.08] | [−0.78, 0.17] |
| | | 12 | 633 | −0.05 | −0.01 | 0.00 | −0.22** | −0.50** | −0.79** |
| | | | | [−0.20, 0.08] | [−0.17, 0.16] | [−0.16, 0.17] | [−0.33, −0.08] | [−0.73, −0.17] | [−1.17, −0.21] |
| | r*-gap | 3 | 642 | 0.06 | 0.00 | −0.04 | −0.08 | −0.16 | −0.27 |
| | | | | [−0.15, 0.18] | [−0.29, 0.12] | [−0.32, 0.10] | [−0.15, 0.17] | [−0.29, 0.38] | [−0.51, 0.54] |
| | | 6 | 639 | 0.09 | 0.05 | 0.05 | −0.14 | −0.35 | −0.61 |
| | | | | [−0.11, 0.23] | [−0.23, 0.22] | [−0.26, 0.21] | [−0.26, 0.04] | [−0.56, 0.08] | [−0.94, 0.10] |
| | | 12 | 633 | 0.18 | 0.14 | 0.13 | −0.27** | −0.63** | −0.98** |
| | | | | [−0.07, 0.38] | [−0.17, 0.36] | [−0.19, 0.35] | [−0.41, −0.10] | [−0.89, −0.22] | [−1.42, −0.26] |

*Notes:* r*-gap only. Pre-1990 months use monthly series in place of the daily data (see lib/model.py); shocks are the 252 announcements of 1996 on. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H12: Section 4.1 (B): 2020 Dropped from All Equations, Start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 359 | 0.13 | 0.22 | 0.26 | −0.18* | −0.28 | −0.35 |
| | | | | [−0.07, 0.29] | [−0.11, 0.47] | [−0.13, 0.53] | [−0.33, 0.00] | [−0.54, 0.05] | [−0.70, 0.12] |

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| | | 6 | 359 | 0.04 | 0.05 | 0.07 | −0.15 | −0.30 | −0.44 |
| | | | | [−0.16, 0.24] | [−0.24, 0.35] | [−0.27, 0.38] | [−0.29, 0.03] | [−0.56, 0.06] | [−0.81, 0.06] |
| | | 12 | 359 | −0.10 | −0.21 | −0.25 | −0.20** | −0.32 | −0.44 |
| | | | | [−0.28, 0.10] | [−0.47, 0.08] | [−0.54, 0.11] | [−0.35, −0.03] | [−0.63, 0.03] | [−0.91, 0.06] |
| | r*-gap | 3 | 359 | 0.37** | 0.44** | 0.39* | −0.22 | −0.35 | −0.47 |
| | | | | [0.14, 0.57] | [0.10, 0.75] | [0.01, 0.82] | [−0.42, 0.01] | [−0.70, 0.07] | [−0.94, 0.11] |
| | | 6 | 359 | 0.38** | 0.44** | 0.43 | −0.30** | −0.57** | −0.80* |
| | | | | [0.15, 0.56] | [0.07, 0.72] | [0.00, 0.75] | [−0.45, −0.08] | [−0.90, −0.09] | [−1.27, −0.08] |
| | | 12 | 359 | 0.41** | 0.45** | 0.40 | −0.38** | −0.69** | −0.94** |
| | | | | [0.17, 0.60] | [0.09, 0.77] | [−0.04, 0.87] | [−0.55, −0.16] | [−1.08, −0.21] | [−1.60, −0.16] |

*Notes:* r*-gap only. March to December 2020 dropped from the monthly and the daily equations. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H13: Section 4.1 (C): 2020 Dropped from All Equations, Start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 632 | 0.06 | 0.11 | 0.13 | −0.11 | −0.20 | −0.27 |
| | | | | [−0.04, 0.16] | [−0.05, 0.25] | [−0.05, 0.30] | [−0.13, 0.15] | [−0.23, 0.40] | [−0.30, 0.64] |
| | | 6 | 629 | 0.11 | 0.10 | 0.05 | −0.16 | −0.39 | −0.62 |
| | | | | [−0.01, 0.21] | [−0.08, 0.23] | [−0.12, 0.20] | [−0.26, 0.01] | [−0.56, 0.03] | [−0.86, 0.07] |
| | | 12 | 623 | 0.17** | 0.15 | 0.05 | −0.28** | −0.64** | −0.97** |
| | | | | [0.04, 0.29] | [−0.03, 0.31] | [−0.14, 0.22] | [−0.37, −0.16] | [−0.83, −0.32] | [−1.27, −0.45] |
| | r*-gap | 3 | 632 | 0.12 | 0.09 | 0.06 | −0.09 | −0.25 | −0.48 |
| | | | | [−0.02, 0.18] | [−0.12, 0.19] | [−0.16, 0.18] | [−0.16, 0.16] | [−0.42, 0.29] | [−0.76, 0.31] |
| | | 6 | 629 | 0.27** | 0.15 | 0.05 | −0.20* | −0.49* | −0.80* |
| | | | | [0.10, 0.36] | [−0.09, 0.27] | [−0.19, 0.19] | [−0.31, −0.01] | [−0.68, −0.03] | [−1.06, −0.06] |
| | | 12 | 623 | 0.33** | 0.24 | 0.15 | −0.32** | −0.76** | −1.14** |
| | | | | [0.14, 0.46] | [−0.02, 0.40] | [−0.09, 0.35] | [−0.42, −0.16] | [−0.93, −0.36] | [−1.46, −0.50] |

*Notes:* r*-gap only. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H14: Section 4.1 (D): Sample Ends in December 2019, Start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 288 | −0.06 | −0.12 | −0.14 | −0.03 | 0.01 | 0.03 |
| | | | | [−0.29, 0.15] | [−0.46, 0.20] | [−0.56, 0.21] | [−0.12, 0.08] | [−0.16, 0.16] | [−0.18, 0.23] |
| | | 6 | 288 | −0.09 | −0.17 | −0.19 | 0.00 | 0.01 | 0.02 |
| | | | | [−0.33, 0.13] | [−0.57, 0.22] | [−0.62, 0.24] | [−0.11, 0.11] | [−0.17, 0.17] | [−0.21, 0.23] |
| | | 12 | 288 | −0.15 | −0.24 | −0.26 | 0.08 | 0.06 | 0.03 |
| | | | | [−0.37, 0.11] | [−0.62, 0.19] | [−0.75, 0.26] | [−0.05, 0.20] | [−0.10, 0.20] | [−0.17, 0.22] |
| | r*-gap | 3 | 288 | 0.24 | 0.19 | 0.11 | −0.11 | −0.10 | −0.08 |

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| | | | | [−0.07, 0.54] | [−0.21, 0.64] | [−0.30, 0.62] | [−0.24, 0.02] | [−0.32, 0.10] | [−0.36, 0.16] |
| | | 6 | 288 | 0.23 | 0.10 | 0.02 | −0.12 | −0.10 | −0.10 |
| | | | | [−0.06, 0.46] | [−0.31, 0.48] | [−0.36, 0.47] | [−0.26, 0.04] | [−0.35, 0.14] | [−0.40, 0.18] |
| | | 12 | 288 | 0.30* | 0.13 | 0.00 | −0.08 | −0.14 | −0.21 |
| | | | | [0.04, 0.55] | [−0.30, 0.49] | [−0.48, 0.42] | [−0.23, 0.08] | [−0.38, 0.09] | [−0.53, 0.09] |

*Notes:* r*-gap only. All equations, monthly and daily. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H15: Section 4.1 (E): Sample Ends in December 2019, Start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 561 | 0.03 | 0.05 | 0.06 | −0.04 | −0.06 | −0.07 |
| | | | | [−0.08, 0.13] | [−0.09, 0.19] | [−0.08, 0.21] | [−0.07, 0.23] | [−0.08, 0.56] | [−0.10, 0.85] |
| | | 6 | 558 | 0.11 | 0.09 | 0.04 | −0.10 | −0.26 | −0.44 |
| | | | | [−0.02, 0.20] | [−0.09, 0.22] | [−0.15, 0.18] | [−0.18, 0.07] | [−0.40, 0.20] | [−0.68, 0.31] |
| | | 12 | 552 | 0.19** | 0.18 | 0.08 | −0.21** | −0.49** | −0.82** |
| | | | | [0.06, 0.31] | [−0.04, 0.32] | [−0.17, 0.25] | [−0.31, −0.06] | [−0.69, −0.15] | [−1.20, −0.25] |
| | r*-gap | 3 | 561 | 0.04 | −0.02 | −0.04 | −0.02 | −0.13 | −0.32 |
| | | | | [−0.11, 0.11] | [−0.22, 0.11] | [−0.26, 0.10] | [−0.07, 0.20] | [−0.30, 0.30] | [−0.63, 0.31] |
| | | 6 | 558 | 0.21* | 0.04 | −0.02 | −0.12 | −0.32 | −0.56 |
| | | | | [0.01, 0.30] | [−0.22, 0.14] | [−0.27, 0.10] | [−0.21, 0.07] | [−0.48, 0.07] | [−0.82, 0.07] |
| | | 12 | 552 | 0.27** | 0.16 | 0.09 | −0.22** | −0.50** | −0.84** |
| | | | | [0.06, 0.41] | [−0.16, 0.35] | [−0.23, 0.28] | [−0.33, −0.06] | [−0.70, −0.16] | [−1.19, −0.23] |

*Notes:* r*-gap only. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H16: Section 4.2 (a): Kiley r*, Start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap (Kiley) | none | 3 | 369 | −0.33** | −0.35** | −0.31** | −0.06 | −0.04 | −0.03 |
| | | | | [−0.57, −0.10] | [−0.61, −0.09] | [−0.59, −0.05] | [−0.21, 0.10] | [−0.32, 0.24] | [−0.41, 0.36] |
| | | 6 | 369 | −0.28** | −0.28* | −0.23 | −0.07 | −0.12 | −0.19 |
| | | | | [−0.57, −0.07] | [−0.56, −0.02] | [−0.53, 0.03] | [−0.21, 0.10] | [−0.37, 0.17] | [−0.58, 0.24] |
| | | 12 | 369 | −0.53** | −0.32* | −0.23 | −0.09 | −0.10 | −0.13 |
| | | | | [−0.86, −0.12] | [−0.62, −0.07] | [−0.55, 0.04] | [−0.24, 0.06] | [−0.37, 0.22] | [−0.55, 0.34] |
| | r*-gap (Kiley) | 3 | 369 | 0.15 | −0.06 | −0.14 | −0.10 | −0.07 | −0.07 |
| | | | | [−0.18, 0.43] | [−0.33, 0.23] | [−0.37, 0.12] | [−0.33, 0.11] | [−0.43, 0.20] | [−0.48, 0.25] |
| | | 6 | 369 | 0.34* | 0.15 | 0.00 | −0.21* | −0.27 | −0.33 |
| | | | | [0.03, 0.66] | [−0.18, 0.50] | [−0.27, 0.32] | [−0.40, −0.01] | [−0.68, 0.08] | [−0.94, 0.12] |
| | | 12 | 369 | 0.39 | 0.28 | −0.01 | −0.25** | −0.18 | −0.03 |
| | | | | [−0.03, 0.79] | [−0.12, 0.69] | [−0.48, 0.36] | [−0.45, −0.05] | [−0.59, 0.21] | [−0.55, 0.56] |

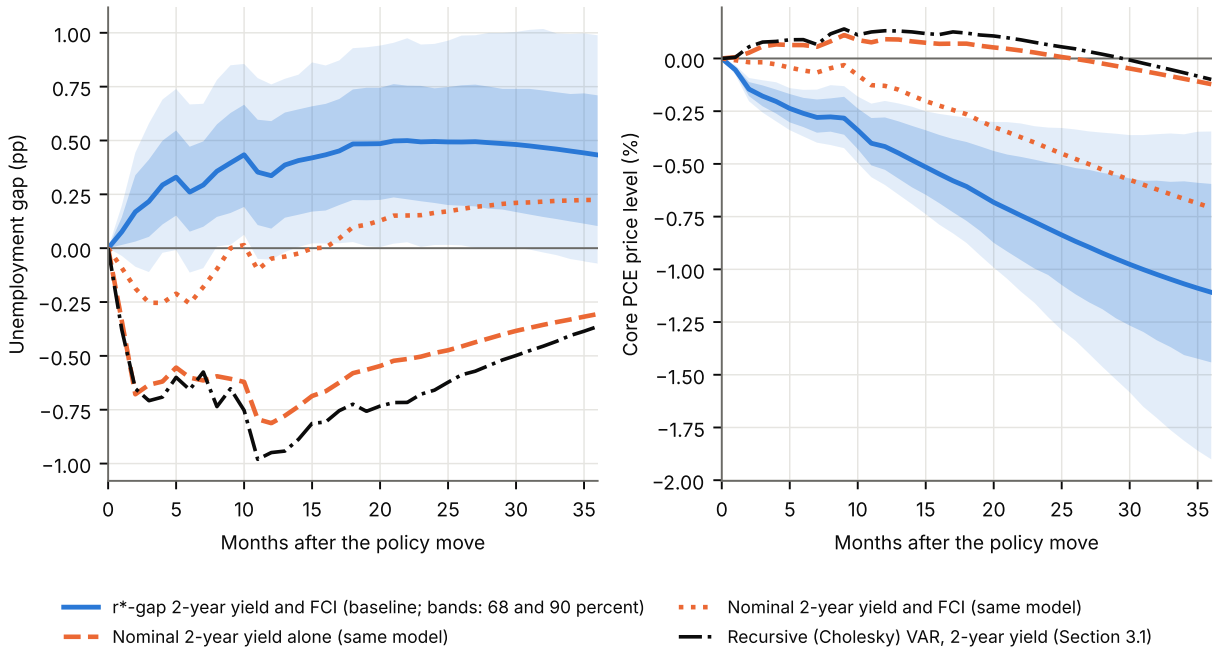



Figure H2: Responses to a Policy Move: Baseline, Nominal Yield Alone and with the FCI, and Cholesky VAR

*Notes:* Policy move that raises the two-year yield 1 percentage point. Baseline: two-year yield and FCI relative to $r^*$; 12 lags; shaded bands 68 and 90 percent. The nominal yield alone and with the FCI are in the same model; the Cholesky VAR is that of Section 3.1.

Table H4: Baseline: 1996 to the Present, All Data in, No Trend (12 Lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | none | 12 | 369 | −0.81** | −0.49** | −0.31 | 0.09 | 0.02 | −0.12 |
| | | | | [−1.20, −0.38] | [−0.82, −0.13] | [−0.69, 0.06] | [−0.11, 0.28] | [−0.33, 0.42] | [−0.61, 0.46] |
| | nominal | 12 | 369 | −0.05 | 0.16 | 0.22 | −0.13 | −0.43 | −0.71 |
| | | | | [−0.50, 0.35] | [−0.24, 0.55] | [−0.20, 0.67] | [−0.33, 0.09] | [−0.80, 0.02] | [−1.22, 0.01] |
| Real | none | 12 | 369 | −0.60** | −0.33* | −0.22 | −0.03 | −0.11 | −0.21 |
| | | | | [−0.96, −0.27] | [−0.64, −0.07] | [−0.55, 0.05] | [−0.18, 0.14] | [−0.37, 0.21] | [−0.61, 0.28] |
| | real | 12 | 369 | 0.09 | 0.23 | 0.21 | −0.23** | −0.48** | −0.66** |
| | | | | [−0.36, 0.43] | [−0.11, 0.61] | [−0.14, 0.62] | [−0.41, −0.06] | [−0.84, −0.12] | [−1.24, −0.10] |
| r*-gap | none | 12 | 369 | −0.29 | −0.07 | 0.03 | −0.21** | −0.44** | −0.69** |
| | | | | [−0.60, 0.11] | [−0.42, 0.24] | [−0.35, 0.36] | [−0.35, −0.05] | [−0.73, −0.05] | [−1.13, −0.10] |
| | r*-gap | 12 | 369 | 0.34 | 0.50* | 0.43 | −0.42** | −0.81** | −1.11** |
| | | | | [−0.05, 0.76] | [0.01, 0.95] | [−0.07, 0.99] | [−0.60, −0.20] | [−1.23, −0.33] | [−1.90, −0.35] |

*Notes:* Nominal, real and r*-gap 2-year yield; the FCI is Note A's index in the matching form in each case. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H5: Section 4.1: Sample (r*-gap 2-Year Yield and FCI) (12 Lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Start 1973 | none | 12 | 633 | −0.05 | −0.01 | 0.00 | −0.22** | −0.50** | −0.79** |
| | | | | [−0.20, 0.08] | [−0.17, 0.16] | [−0.16, 0.17] | [−0.33, −0.08] | [−0.73, −0.17] | [−1.17, −0.21] |
| | r*-gap | 12 | 633 | 0.18 | 0.14 | 0.13 | −0.27** | −0.63** | −0.98** |
| | | | | [−0.07, 0.38] | [−0.17, 0.36] | [−0.19, 0.35] | [−0.41, −0.10] | [−0.89, −0.22] | [−1.42, −0.26] |
| 2020 dropped | none | 12 | 359 | −0.10 | −0.21 | −0.25 | −0.20** | −0.32 | −0.44 |
| | | | | [−0.28, 0.10] | [−0.47, 0.08] | [−0.54, 0.11] | [−0.35, −0.03] | [−0.63, 0.03] | [−0.91, 0.06] |
| | r*-gap | 12 | 359 | 0.41** | 0.45** | 0.40 | −0.38** | −0.69** | −0.94** |
| | | | | [0.17, 0.60] | [0.09, 0.77] | [−0.04, 0.87] | [−0.55, −0.16] | [−1.08, −0.21] | [−1.60, −0.16] |
| 2020 dropped, start 1973 | none | 12 | 623 | 0.17** | 0.15 | 0.05 | −0.28** | −0.64** | −0.97** |
| | | | | [0.04, 0.29] | [−0.03, 0.31] | [−0.14, 0.22] | [−0.37, −0.16] | [−0.83, −0.32] | [−1.27, −0.45] |
| | r*-gap | 12 | 623 | 0.33** | 0.24 | 0.15 | −0.32** | −0.76** | −1.14** |
| | | | | [0.14, 0.46] | [−0.02, 0.40] | [−0.09, 0.35] | [−0.42, −0.16] | [−0.93, −0.36] | [−1.46, −0.50] |
| Ends 2019 | none | 12 | 288 | −0.15 | −0.24 | −0.26 | 0.08 | 0.06 | 0.03 |
| | | | | [−0.37, 0.11] | [−0.62, 0.19] | [−0.75, 0.26] | [−0.05, 0.20] | [−0.10, 0.20] | [−0.17, 0.22] |
| | r*-gap | 12 | 288 | 0.30* | 0.13 | 0.00 | −0.08 | −0.14 | −0.21 |
| | | | | [0.04, 0.55] | [−0.30, 0.49] | [−0.48, 0.42] | [−0.23, 0.08] | [−0.38, 0.09] | [−0.53, 0.09] |
| Ends 2019, start 1973 | none | 12 | 552 | 0.19** | 0.18 | 0.08 | −0.21** | −0.49** | −0.82** |
| | | | | [0.06, 0.31] | [−0.04, 0.32] | [−0.17, 0.25] | [−0.31, −0.06] | [−0.69, −0.15] | [−1.20, −0.25] |
| | r*-gap | 12 | 552 | 0.27** | 0.16 | 0.09 | −0.22** | −0.50** | −0.84** |
| | | | | [0.06, 0.41] | [−0.16, 0.35] | [−0.23, 0.28] | [−0.33, −0.06] | [−0.70, −0.16] | [−1.19, −0.23] |

*Notes:* Start 1973: monthly equations from 1973, with monthly series in place of daily data before 1990 (lib/model.py). 2020 dropped: March to December 2020 removed from the monthly and daily equations (2020 values still enter as lags). Ends 2019: all equations, monthly and daily. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

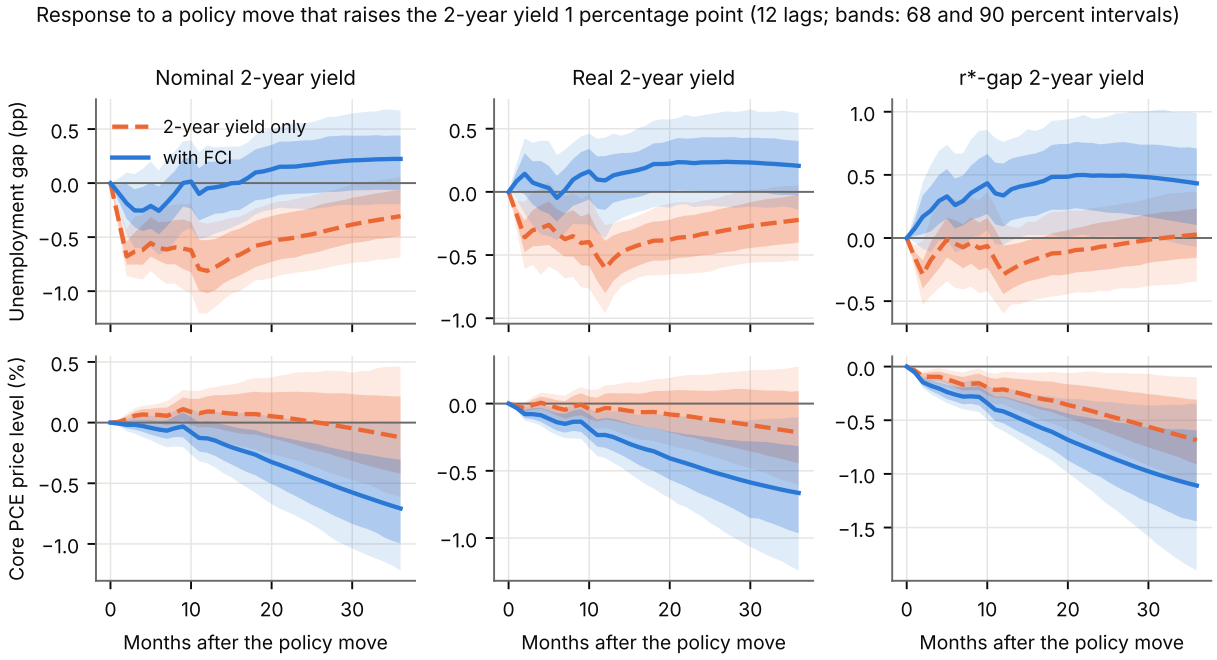



Figure H3: Responses to a Policy Move, without and with the FCI

*Notes:* Policy move that raises the two-year yield 1 percentage point; 12 lags. Bands: 68 and 90 percent intervals.

Table H6: Section 4.2: the r* Adjustment and Trends (12 Lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Kiley r* | none | 12 | 369 | $-0.53^{**}$ | $-0.32^{*}$ | $-0.23$ | $-0.09$ | $-0.10$ | $-0.13$ |
| | | | | [−0.86, −0.12] | [−0.62, −0.07] | [−0.55, 0.04] | [−0.24, 0.06] | [−0.37, 0.22] | [−0.55, 0.34] |
| | r*-gap (Kiley) | 12 | 369 | 0.39 | 0.28 | $-0.01$ | $-0.25^{**}$ | $-0.18$ | $-0.03$ |
| | | | | [−0.03, 0.79] | [−0.12, 0.69] | [−0.48, 0.36] | [−0.45, −0.05] | [−0.59, 0.21] | [−0.55, 0.56] |
| Kiley r*, start 1973 | none | 12 | 633 | $-0.11$ | $-0.05$ | $-0.03$ | $-0.18^{**}$ | $-0.35^{**}$ | $-0.50^{*}$ |
| | | | | [−0.24, 0.02] | [−0.19, 0.07] | [−0.17, 0.10] | [−0.26, −0.04] | [−0.52, −0.07] | [−0.79, −0.04] |
| | r*-gap (Kiley) | 12 | 633 | 0.00 | $-0.04$ | $-0.03$ | $-0.20^{**}$ | $-0.37^{**}$ | $-0.50^{**}$ |
| | | | | [−0.18, 0.17] | [−0.24, 0.11] | [−0.21, 0.11] | [−0.28, −0.07] | [−0.55, −0.11] | [−0.79, −0.07] |
| Real, spline trend | none | 12 | 369 | $-0.64^{**}$ | $-0.34^{*}$ | $-0.20$ | 0.03 | $-0.06$ | $-0.21$ |
| | | | | [−1.02, −0.27] | [−0.70, −0.01] | [−0.55, 0.10] | [−0.13, 0.20] | [−0.33, 0.26] | [−0.61, 0.26] |
| | real | 12 | 369 | 0.19 | 0.28 | 0.17 | $-0.14$ | $-0.25$ | $-0.27$ |
| | | | | [−0.24, 0.55] | [−0.20, 0.66] | [−0.25, 0.52] | [−0.30, 0.07] | [−0.66, 0.15] | [−0.91, 0.26] |
| Real, spline trend, start 1973 | none | 12 | 633 | $-0.10$ | 0.08 | 0.11 | $-0.09$ | $-0.25^{**}$ | $-0.38^{**}$ |
| | | | | [−0.26, 0.04] | [−0.15, 0.28] | [−0.12, 0.25] | [−0.17, 0.01] | [−0.41, −0.04] | [−0.61, −0.07] |
| | real | 12 | 633 | 0.18 | 0.18 | 0.11 | $-0.17^{**}$ | $-0.39^{**}$ | $-0.53^{**}$ |
| | | | | [−0.03, 0.39] | [−0.11, 0.42] | [−0.17, 0.31] | [−0.26, −0.04] | [−0.59, −0.14] | [−0.79, −0.17] |

*Notes:* Kiley r*: 2-year minus expected inflation minus Kiley's r*, with Note A's index built with the same r*. Real, spline trend: real 2-year and real FCI with a natural cubic spline trend (knots at the sample ends and at NBER business-cycle peaks inside the sample) in every monthly equation, in place of any r* measure. Per +1 pp on the 2-year yield. $^{**}$ and $^{*}$: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H7: Section 4.3: Excess Bond Premium (r*-gap) (12 Lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| EBP added | none | 12 | 369 | −0.08 | 0.08 | 0.16 | −0.27** | −0.53** | −0.78** |
| | | | | [−0.40, 0.25] | [−0.27, 0.36] | [−0.18, 0.44] | [−0.39, −0.08] | [−0.80, −0.17] | [−1.21, −0.24] |
| | r*-gap | 12 | 369 | 0.27 | 0.52* | 0.42 | −0.39** | −0.78** | −1.10** |
| | | | | [−0.17, 0.63] | [0.04, 0.99] | [−0.10, 1.03] | [−0.56, −0.17] | [−1.24, −0.31] | [−1.93, −0.32] |
| EBP added, start 1973 | none | 12 | 633 | −0.03 | −0.01 | 0.01 | −0.22** | −0.50** | −0.78** |
| | | | | [−0.17, 0.09] | [−0.17, 0.12] | [−0.15, 0.15] | [−0.31, −0.07] | [−0.67, −0.16] | [−1.10, −0.24] |
| | r*-gap | 12 | 633 | 0.06 | 0.03 | 0.04 | −0.24** | −0.57** | −0.89** |
| | | | | [−0.14, 0.23] | [−0.18, 0.24] | [−0.15, 0.23] | [−0.35, −0.06] | [−0.77, −0.16] | [−1.22, −0.22] |

*Notes:* The excess bond premium is a third monthly variable, treated like unemployment and inflation (u, inflation, EBP system); responses shown for u and the price level. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H8: Section 3.6: the FCI Replaced by Its Three Weighted Components (12 Lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | components | 12 | 369 | 0.03 | 0.25 | 0.09 | 0.03 | 0.21 | 0.42 |
| | | | | [−0.55, 0.48] | [−0.29, 0.83] | [−0.52, 0.93] | [−0.19, 0.27] | [−0.23, 0.66] | [−0.25, 1.02] |
| Real | components | 12 | 369 | 0.17 | 0.29 | −0.03 | −0.06 | 0.14 | 0.41 |
| | | | | [−0.38, 0.52] | [−0.19, 0.75] | [−0.53, 0.63] | [−0.23, 0.13] | [−0.27, 0.48] | [−0.21, 0.92] |
| r*-gap | components | 12 | 369 | 0.20 | 0.22 | −0.09 | −0.14 | 0.06 | 0.35 |
| | | | | [−0.31, 0.65] | [−0.26, 0.67] | [−0.67, 0.45] | [−0.29, 0.04] | [−0.25, 0.43] | [−0.25, 0.87] |
| Nominal, start 1973 | components | 12 | 633 | −0.28 | −0.24 | −0.26 | 0.10 | 0.45* | 0.92** |
| | | | | [−0.77, 0.09] | [−0.78, 0.17] | [−0.89, 0.28] | [−0.08, 0.37] | [0.08, 1.13] | [0.25, 2.07] |
| Real, start 1973 | components | 12 | 633 | −0.11 | −0.14 | −0.15 | −0.14 | −0.05 | 0.16 |
| | | | | [−0.51, 0.17] | [−0.61, 0.13] | [−0.63, 0.12] | [−0.31, 0.15] | [−0.36, 0.70] | [−0.34, 1.39] |
| r*-gap, start 1973 | components | 12 | 633 | 0.17 | 0.16 | 0.09 | −0.35** | −0.60 | −0.80 |
| | | | | [−0.30, 0.52] | [−0.27, 0.50] | [−0.31, 0.38] | [−0.56, −0.10] | [−0.96, 0.06] | [−1.42, 0.32] |

*Notes:* Components: the 10-year yield, equity valuation and real dollar, each multiplied by its weight, enter separately in the daily and monthly equations in the same version as the 2-year yield, each with its own window impact (the three impacts sum to the FCI's). Baseline rows without the FCI are in the baseline table. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H9: Section 4.4: Alternative Surprises (r*-gap) (12 Lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Bauer-Swanson surprise | none | 12 | 369 | −0.29 | −0.07 | 0.03 | −0.21** | −0.44** | −0.69** |
| | | | | [−0.60, 0.11] | [−0.42, 0.24] | [−0.35, 0.36] | [−0.35, −0.05] | [−0.73, −0.05] | [−1.13, −0.10] |
| | r*-gap | 12 | 369 | 0.40 | 0.54* | 0.47 | −0.44** | −0.84** | −1.15** |
| | | | | [−0.05, 0.88] | [0.06, 1.05] | [−0.04, 1.08] | [−0.64, −0.22] | [−1.29, −0.34] | [−2.00, −0.36] |
| Jarocinski-Karadi surprise | none | 12 | 369 | −0.29 | −0.07 | 0.03 | −0.21** | −0.44** | −0.68** |
| | | | | [−0.60, 0.11] | [−0.42, 0.24] | [−0.35, 0.36] | [−0.35, −0.05] | [−0.72, −0.05] | [−1.13, −0.10] |
| | r*-gap | 12 | 369 | 0.42* | 0.56* | 0.49 | −0.45** | −0.85** | −1.17** |
| | | | | [0.00, 0.88] | [0.05, 1.06] | [−0.04, 1.05] | [−0.64, −0.22] | [−1.29, −0.36] | [−2.01, −0.34] |

*Notes:* Bauer-Swanson: orthogonalized surprise of Bauer and Swanson (2023), announcements through 2019. Jarocinski-Karadi: Nakamura-Steinsson surprise kept when the 2-year yield and the S&P 500 move in opposite directions in the window. The window impact and the bootstrap use the events with a surprise. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

*Notes:* 2-year minus expected inflation minus Kiley's r*; the FCI is Note A's index built with Kiley's r* (same adjustment in rate and FCI). Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H17: Section 4.2 (B): Kiley r*, Start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap (Kiley) | none | 3 | 642 | −0.02 | 0.00 | 0.00 | −0.04 | −0.02 | 0.01* |
| | | | | [−0.12, 0.09] | [−0.12, 0.12] | [−0.10, 0.11] | [−0.06, 0.20] | [−0.03, 0.49] | [0.01, 0.78] |
| | | 6 | 639 | −0.04 | −0.01 | −0.01 | −0.07 | −0.17 | −0.28 |
| | | | | [−0.15, 0.07] | [−0.14, 0.12] | [−0.12, 0.12] | [−0.16, 0.09] | [−0.32, 0.21] | [−0.50, 0.36] |
| | | 12 | 633 | −0.11 | −0.05 | −0.03 | −0.18** | −0.35** | −0.50* |
| | | | | [−0.24, 0.02] | [−0.19, 0.07] | [−0.17, 0.10] | [−0.26, −0.04] | [−0.52, −0.07] | [−0.79, −0.04] |
| | r*-gap (Kiley) | 3 | 642 | −0.07 | −0.11* | −0.11 | −0.05 | −0.11 | −0.18 |
| | | | | [−0.22, 0.02] | [−0.28, −0.01] | [−0.26, 0.00] | [−0.11, 0.12] | [−0.26, 0.25] | [−0.41, 0.31] |
| | | 6 | 639 | −0.06 | −0.09 | −0.07 | −0.08 | −0.18 | −0.30 |
| | | | | [−0.24, 0.05] | [−0.29, 0.05] | [−0.28, 0.05] | [−0.16, 0.08] | [−0.35, 0.13] | [−0.60, 0.20] |
| | | 12 | 633 | 0.00 | −0.04 | −0.03 | −0.20** | −0.37** | −0.50** |
| | | | | [−0.18, 0.17] | [−0.24, 0.11] | [−0.21, 0.11] | [−0.28, −0.07] | [−0.55, −0.11] | [−0.79, −0.07] |

*Notes:* As 2a. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H18: Section 4.2 (C): Real 2-Year and Real FCI with a Cubic Spline Trend, Start 1996

| | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|
| Case FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Real none | 3 | 369 | −0.44** | −0.40** | −0.34** | −0.07 | −0.14 | −0.21 |
| | | | [−0.71, −0.17] | [−0.69, −0.10] | [−0.69, −0.06] | [−0.24, 0.08] | [−0.45, 0.12] | [−0.62, 0.19] |
| | 6 | 369 | −0.44** | −0.34** | −0.24 | −0.03 | −0.21 | −0.41 |
| | | | [−0.77, −0.20] | [−0.67, −0.05] | [−0.62, 0.01] | [−0.16, 0.13] | [−0.49, 0.11] | [−0.76, 0.08] |
| | 12 | 369 | −0.64** | −0.34* | −0.20 | 0.03 | −0.06 | −0.21 |
| | | | [−1.02, −0.27] | [−0.70, −0.01] | [−0.55, 0.10] | [−0.13, 0.20] | [−0.33, 0.26] | [−0.61, 0.26] |
| real | 3 | 369 | 0.00 | −0.03 | −0.05 | −0.17 | −0.29 | −0.36 |
| | | | [−0.40, 0.28] | [−0.41, 0.22] | [−0.40, 0.17] | [−0.35, 0.03] | [−0.59, 0.05] | [−0.76, 0.06] |
| | 6 | 369 | 0.11 | 0.20 | 0.15 | −0.20* | −0.49** | −0.68** |
| | | | [−0.33, 0.42] | [−0.25, 0.46] | [−0.23, 0.38] | [−0.38, 0.00] | [−0.78, −0.10] | [−1.05, −0.15] |
| | 12 | 369 | 0.19 | 0.28 | 0.17 | −0.14 | −0.25 | −0.27 |
| | | | [−0.24, 0.55] | [−0.20, 0.66] | [−0.25, 0.52] | [−0.30, 0.07] | [−0.66, 0.15] | [−0.91, 0.26] |

*Notes:* No r* measure: the trend is a natural cubic spline in time (knots at the sample ends and at the NBER business-cycle peaks inside the sample), estimated in every monthly equation. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H19: Section 4.2 (D): Real 2-Year and Real FCI with a Cubic Spline Trend, Start 1973

| | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|
| Case FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Real none | 3 | 642 | 0.01 | 0.04 | 0.04 | −0.11* | −0.22** | −0.29** |
| | | | [−0.15, 0.16] | [−0.14, 0.19] | [−0.11, 0.15] | [−0.21, −0.01] | [−0.38, −0.03] | [−0.51, −0.05] |
| | 6 | 639 | 0.01 | 0.06 | 0.06 | −0.11 | −0.31** | −0.47** |
| | | | [−0.13, 0.19] | [−0.10, 0.24] | [−0.09, 0.17] | [−0.19, 0.00] | [−0.45, −0.07] | [−0.66, −0.12] |
| | 12 | 633 | −0.10 | 0.08 | 0.11 | −0.09 | −0.25** | −0.38** |
| | | | [−0.26, 0.04] | [−0.15, 0.28] | [−0.12, 0.25] | [−0.17, 0.01] | [−0.41, −0.04] | [−0.61, −0.07] |
| real | 3 | 642 | 0.22 | 0.15 | 0.06 | −0.17** | −0.31** | −0.38** |
| | | | [−0.01, 0.42] | [−0.10, 0.40] | [−0.16, 0.32] | [−0.30, −0.04] | [−0.51, −0.09] | [−0.65, −0.10] |
| | 6 | 639 | 0.23** | 0.21 | 0.11 | −0.17** | −0.43** | −0.61** |
| | | | [0.04, 0.42] | [−0.04, 0.43] | [−0.12, 0.33] | [−0.27, −0.03] | [−0.61, −0.13] | [−0.85, −0.20] |
| | 12 | 633 | 0.18 | 0.18 | 0.11 | −0.17** | −0.39** | −0.53** |
| | | | [−0.03, 0.39] | [−0.11, 0.42] | [−0.17, 0.31] | [−0.26, −0.04] | [−0.59, −0.14] | [−0.79, −0.17] |

*Notes:* As 2c; knots at the NBER peaks of 1973, 1980, 1981, 1990, 2001, 2007 and 2020. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H20: Section 4.3 (a): Excess Bond Premium as a Third Monthly Variable, Start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 369 | −0.11 | −0.14 | −0.14 | −0.14 | −0.17 | −0.19 |
| | | | | [−0.40, 0.19] | [−0.43, 0.18] | [−0.41, 0.13] | [−0.28, 0.02] | [−0.43, 0.14] | [−0.52, 0.25] |
| | | 6 | 369 | 0.02 | −0.05 | −0.09 | −0.19** | −0.36* | −0.53* |
| | | | | [−0.24, 0.29] | [−0.36, 0.22] | [−0.42, 0.17] | [−0.34, −0.03] | [−0.66, −0.05] | [−0.95, −0.04] |
| | | 12 | 369 | −0.08 | 0.08 | 0.16 | −0.27** | −0.53** | −0.78** |
| | | | | [−0.40, 0.25] | [−0.27, 0.36] | [−0.18, 0.44] | [−0.39, −0.08] | [−0.80, −0.17] | [−1.21, −0.24] |
| | r*-gap | 3 | 369 | 0.19 | 0.03 | −0.14 | −0.18 | −0.20 | −0.19 |
| | | | | [−0.12, 0.50] | [−0.38, 0.38] | [−0.53, 0.26] | [−0.39, 0.03] | [−0.53, 0.23] | [−0.59, 0.36] |
| | | 6 | 369 | 0.31 | 0.22 | 0.04 | −0.29** | −0.55** | −0.76** |
| | | | | [−0.02, 0.61] | [−0.20, 0.59] | [−0.40, 0.44] | [−0.47, −0.09] | [−0.90, −0.12] | [−1.33, −0.14] |
| | | 12 | 369 | 0.27 | 0.52* | 0.42 | −0.39** | −0.78** | −1.10** |
| | | | | [−0.17, 0.63] | [0.04, 0.99] | [−0.10, 1.03] | [−0.56, −0.17] | [−1.24, −0.31] | [−1.93, −0.32] |

*Notes:* r*-gap. Monthly system u, inflation, EBP (EBP treated like u and inflation; responses shown for u and the price level). Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H21: Section 4.3 (B): Excess Bond Premium as a Third Monthly Variable, Start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 642 | 0.00 | 0.04 | 0.04 | −0.09 | −0.15 | −0.19 |
| | | | | [−0.10, 0.12] | [−0.09, 0.16] | [−0.07, 0.15] | [−0.12, 0.16] | [−0.16, 0.43] | [−0.20, 0.71] |
| | | 6 | 639 | 0.01 | 0.03 | 0.01 | −0.12 | −0.31 | −0.51 |
| | | | | [−0.11, 0.10] | [−0.12, 0.13] | [−0.13, 0.11] | [−0.21, 0.04] | [−0.44, 0.10] | [−0.70, 0.19] |
| | | 12 | 633 | −0.03 | −0.01 | 0.01 | −0.22** | −0.50** | −0.78** |
| | | | | [−0.17, 0.09] | [−0.17, 0.12] | [−0.15, 0.15] | [−0.31, −0.07] | [−0.67, −0.16] | [−1.10, −0.24] |
| | r*-gap | 3 | 642 | 0.03 | 0.01 | −0.02 | −0.10 | −0.20 | −0.33 |
| | | | | [−0.14, 0.11] | [−0.24, 0.14] | [−0.24, 0.13] | [−0.15, 0.15] | [−0.29, 0.33] | [−0.53, 0.53] |
| | | 6 | 639 | 0.04 | 0.02 | 0.00 | −0.12 | −0.32 | −0.58 |
| | | | | [−0.13, 0.14] | [−0.22, 0.16] | [−0.23, 0.14] | [−0.21, 0.08] | [−0.48, 0.12] | [−0.82, 0.18] |
| | | 12 | 633 | 0.06 | 0.03 | 0.04 | −0.24** | −0.57** | −0.89** |
| | | | | [−0.14, 0.23] | [−0.18, 0.24] | [−0.15, 0.23] | [−0.35, −0.06] | [−0.77, −0.16] | [−1.22, −0.22] |

*Notes:* As 3a. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H22: Section 3.6: the FCI Replaced by Its Three Weighted Components

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | components | 3 | 369 | −0.08 | −0.04 | −0.04 | 0.07 | 0.11 | 0.09 |
| | | | | [−0.47, 0.27] | [−0.42, 0.32] | [−0.41, 0.36] | [−0.17, 0.28] | [−0.30, 0.48] | [−0.41, 0.61] |
| | | 6 | 369 | 0.08 | 0.23 | 0.12 | −0.11 | −0.20 | −0.26 |

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| | | | | [−0.32, 0.47] | [−0.23, 0.69] | [−0.28, 0.66] | [−0.35, 0.14] | [−0.64, 0.22] | [−0.87, 0.32] |
| | | 12 | 369 | 0.03 | 0.25 | 0.09 | 0.03 | 0.21 | 0.42 |
| | | | | [−0.55, 0.48] | [−0.29, 0.83] | [−0.52, 0.93] | [−0.19, 0.27] | [−0.23, 0.66] | [−0.25, 1.02] |
| Real | components | 3 | 369 | 0.09 | −0.01 | −0.06 | −0.03 | 0.00 | −0.02 |
| | | | | [−0.22, 0.38] | [−0.29, 0.23] | [−0.34, 0.25] | [−0.21, 0.15] | [−0.30, 0.30] | [−0.38, 0.36] |
| | | 6 | 369 | 0.25 | 0.21 | 0.01 | −0.11 | −0.14 | −0.14 |
| | | | | [−0.10, 0.54] | [−0.14, 0.59] | [−0.32, 0.46] | [−0.29, 0.08] | [−0.50, 0.19] | [−0.64, 0.27] |
| | | 12 | 369 | 0.17 | 0.29 | −0.03 | −0.06 | 0.14 | 0.41 |
| | | | | [−0.38, 0.52] | [−0.19, 0.75] | [−0.53, 0.63] | [−0.23, 0.13] | [−0.27, 0.48] | [−0.21, 0.92] |
| r*-gap | components | 3 | 369 | 0.05 | −0.10 | −0.16 | 0.06 | 0.23 | 0.29 |
| | | | | [−0.26, 0.36] | [−0.39, 0.21] | [−0.46, 0.18] | [−0.12, 0.28] | [−0.14, 0.58] | [−0.20, 0.75] |
| | | 6 | 369 | 0.23 | 0.12 | −0.05 | −0.13 | −0.10 | −0.07 |
| | | | | [−0.16, 0.52] | [−0.26, 0.43] | [−0.41, 0.34] | [−0.28, 0.10] | [−0.43, 0.34] | [−0.56, 0.53] |
| | | 12 | 369 | 0.20 | 0.22 | −0.09 | −0.14 | 0.06 | 0.35 |
| | | | | [−0.31, 0.65] | [−0.26, 0.67] | [−0.67, 0.45] | [−0.29, 0.04] | [−0.25, 0.43] | [−0.25, 0.87] |

*Notes:* The 10-year yield, equity valuation and the real dollar, each multiplied by its weight, enter the daily and monthly equations separately, in the same version as the 2-year yield; each has its own window impact. Replaces the FCI, so the same step without the FCI is the baseline row. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H23: Section 3.6 (C): the Components with Common Daily Dynamics, 1996 and 1973 Starts

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | index | 3 | 369 | −0.04 | 0.10 | 0.09 | −0.10 | −0.31 | −0.46 |
| | | | | [−0.38, 0.29] | [−0.31, 0.47] | [−0.30, 0.52] | [−0.34, 0.11] | [−0.68, 0.07] | [−0.93, 0.08] |
| | | 6 | 369 | −0.01 | 0.21 | 0.23 | −0.15 | −0.45 | −0.72* |
| | | | | [−0.36, 0.38] | [−0.18, 0.63] | [−0.20, 0.72] | [−0.39, 0.10] | [−0.93, 0.01] | [−1.34, −0.06] |
| | | 12 | 369 | −0.09 | 0.20 | 0.29 | −0.14 | −0.50 | −0.85* |
| | | | | [−0.56, 0.36] | [−0.24, 0.66] | [−0.17, 0.80] | [−0.37, 0.11] | [−0.91, 0.01] | [−1.48, −0.11] |
| | equities free | 3 | 369 | −0.05 | 0.07 | 0.06 | −0.09 | −0.28 | −0.41 |
| | | | | [−0.38, 0.26] | [−0.30, 0.43] | [−0.30, 0.46] | [−0.31, 0.12] | [−0.62, 0.06] | [−0.86, 0.08] |
| | | 6 | 369 | −0.05 | 0.19 | 0.18 | −0.13 | −0.41 | −0.64 |
| | | | | [−0.37, 0.29] | [−0.19, 0.58] | [−0.24, 0.58] | [−0.37, 0.13] | [−0.85, 0.04] | [−1.27, 0.01] |
| | | 12 | 369 | −0.18 | 0.14 | 0.22 | −0.10 | −0.39 | −0.70 |
| | | | | [−0.67, 0.24] | [−0.30, 0.56] | [−0.25, 0.72] | [−0.28, 0.16] | [−0.80, 0.13] | [−1.34, 0.09] |
| | index | 3 | 642 | 0.11 | 0.10 | 0.05 | 0.20** | 0.57** | 0.95** |
| | | | | [−0.12, 0.28] | [−0.20, 0.32] | [−0.31, 0.34] | [0.13, 0.51] | [0.47, 1.20] | [0.76, 1.90] |
| | | 6 | 639 | 0.02 | 0.07 | 0.07 | 0.07 | 0.23* | 0.42* |
| | | | | [−0.26, 0.18] | [−0.25, 0.29] | [−0.28, 0.31] | [−0.08, 0.35] | [0.01, 0.87] | [0.09, 1.47] |
| | | 12 | 633 | 0.05 | 0.03 | 0.01 | −0.01 | 0.01 | 0.07 |
| | | | | [−0.32, 0.32] | [−0.32, 0.27] | [−0.37, 0.24] | [−0.18, 0.23] | [−0.30, 0.56] | [−0.38, 0.96] |
| | equities free | 3 | 642 | −0.17 | −0.20 | −0.22 | 0.18** | 0.48** | 0.80** |
| | | | | [−0.49, 0.02] | [−0.63, 0.10] | [−0.70, 0.17] | [0.06, 0.51] | [0.25, 1.20] | [0.43, 1.91] |
| | | 6 | 639 | −0.26** | −0.18 | −0.15 | 0.09 | 0.24 | 0.43 |
| | | | | [−0.61, −0.06] | [−0.58, 0.10] | [−0.65, 0.21] | [−0.10, 0.38] | [−0.12, 0.95] | [−0.15, 1.64] |

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| | | 12 | 633 | −0.36* | −0.24 | −0.17 | 0.04 | 0.11 | 0.19 |
| | | | | [−0.78, −0.04] | [−0.73, 0.08] | [−0.74, 0.24] | [−0.17, 0.32] | [−0.29, 0.78] | [−0.48, 1.35] |
| Real | index | 3 | 369 | 0.15 | 0.10 | 0.03 | −0.21* | −0.35 | −0.41 |
| | | | | [−0.13, 0.41] | [−0.19, 0.40] | [−0.26, 0.41] | [−0.40, −0.01] | [−0.65, 0.01] | [−0.81, 0.02] |
| | | 6 | 369 | 0.17 | 0.24 | 0.17 | −0.22* | −0.47** | −0.66** |
| | | | | [−0.13, 0.46] | [−0.09, 0.62] | [−0.17, 0.54] | [−0.42, −0.02] | [−0.84, −0.10] | [−1.16, −0.13] |
| | | 12 | 369 | 0.12 | 0.28 | 0.23 | −0.25** | −0.53** | −0.72** |
| | | | | [−0.34, 0.48] | [−0.08, 0.68] | [−0.13, 0.68] | [−0.42, −0.06] | [−0.93, −0.14] | [−1.34, −0.14] |
| | equities free | 3 | 369 | 0.07 | 0.03 | −0.02 | −0.15 | −0.24 | −0.28 |
| | | | | [−0.19, 0.34] | [−0.25, 0.31] | [−0.30, 0.27] | [−0.33, 0.04] | [−0.53, 0.09] | [−0.69, 0.14] |
| | | 6 | 369 | 0.08 | 0.16 | 0.09 | −0.14 | −0.31 | −0.43 |
| | | | | [−0.22, 0.36] | [−0.18, 0.47] | [−0.24, 0.44] | [−0.32, 0.06] | [−0.64, 0.07] | [−0.94, 0.13] |
| | | 12 | 369 | −0.07 | 0.16 | 0.15 | −0.15 | −0.34 | −0.49 |
| | | | | [−0.54, 0.29] | [−0.24, 0.52] | [−0.24, 0.58] | [−0.32, 0.07] | [−0.70, 0.08] | [−1.05, 0.17] |
| | index | 3 | 642 | 0.21 | 0.11 | 0.02 | 0.08** | 0.32** | 0.49** |
| | | | | [−0.05, 0.33] | [−0.20, 0.30] | [−0.28, 0.27] | [0.08, 0.43] | [0.33, 1.08] | [0.36, 1.63] |
| | | 6 | 639 | 0.16 | 0.15 | 0.09 | −0.10 | −0.12 | −0.10 |
| | | | | [−0.11, 0.24] | [−0.17, 0.26] | [−0.22, 0.23] | [−0.20, 0.16] | [−0.22, 0.49] | [−0.20, 0.93] |
| | | 12 | 633 | 0.24 | 0.12 | 0.05 | −0.26** | −0.45* | −0.57 |
| | | | | [−0.07, 0.46] | [−0.19, 0.27] | [−0.24, 0.21] | [−0.41, −0.06] | [−0.71, −0.03] | [−0.92, 0.11] |
| | equities free | 3 | 642 | −0.05 | −0.15 | −0.20 | 0.16** | 0.52** | 0.85** |
| | | | | [−0.38, 0.09] | [−0.57, 0.08] | [−0.70, 0.11] | [0.11, 0.55] | [0.41, 1.35] | [0.51, 2.18] |
| | | 6 | 639 | −0.11 | −0.09 | −0.09 | −0.02 | 0.05 | 0.17 |
| | | | | [−0.47, 0.03] | [−0.52, 0.08] | [−0.60, 0.11] | [−0.16, 0.26] | [−0.18, 0.80] | [−0.17, 1.45] |
| | | 12 | 633 | −0.23 | −0.18 | −0.14 | −0.13 | −0.19 | −0.22 |
| | | | | [−0.61, 0.04] | [−0.63, 0.07] | [−0.64, 0.11] | [−0.33, 0.11] | [−0.53, 0.40] | [−0.76, 0.82] |
| r*-gap | index | 3 | 369 | 0.24 | 0.10 | −0.03 | −0.18 | −0.22 | −0.24 |
| | | | | [−0.06, 0.55] | [−0.23, 0.41] | [−0.34, 0.32] | [−0.37, 0.03] | [−0.54, 0.14] | [−0.65, 0.22] |
| | | 6 | 369 | 0.32 | 0.29 | 0.17 | −0.30** | −0.56** | −0.76** |
| | | | | [−0.01, 0.60] | [−0.09, 0.65] | [−0.23, 0.55] | [−0.48, −0.10] | [−0.92, −0.15] | [−1.28, −0.18] |
| | | 12 | 369 | 0.30 | 0.40* | 0.30 | −0.38** | −0.72** | −0.97** |
| | | | | [−0.09, 0.68] | [0.03, 0.86] | [−0.12, 0.84] | [−0.55, −0.19] | [−1.11, −0.31] | [−1.62, −0.32] |
| | equities free | 3 | 369 | 0.13 | 0.00 | −0.11 | −0.09 | −0.04 | 0.00 |
| | | | | [−0.16, 0.45] | [−0.30, 0.30] | [−0.38, 0.22] | [−0.27, 0.14] | [−0.38, 0.37] | [−0.45, 0.52] |
| | | 6 | 369 | 0.22 | 0.20 | 0.08 | −0.22 | −0.39 | −0.51 |
| | | | | [−0.12, 0.51] | [−0.17, 0.50] | [−0.28, 0.40] | [−0.38, 0.00] | [−0.74, 0.07] | [−1.04, 0.15] |
| | | 12 | 369 | 0.17 | 0.33 | 0.22 | −0.29** | −0.52** | −0.66* |
| | | | | [−0.30, 0.56] | [−0.06, 0.71] | [−0.21, 0.69] | [−0.45, −0.08] | [−0.89, −0.09] | [−1.29, −0.02] |
| | index | 3 | 642 | 0.34* | 0.20 | 0.04 | −0.01 | 0.20** | 0.29** |
| | | | | [0.03, 0.50] | [−0.21, 0.45] | [−0.31, 0.35] | [0.00, 0.40] | [0.28, 1.02] | [0.17, 1.57] |
| | | 6 | 639 | 0.34** | 0.26 | 0.13 | −0.22 | −0.30 | −0.38 |
| | | | | [0.07, 0.52] | [−0.06, 0.49] | [−0.18, 0.35] | [−0.32, 0.04] | [−0.44, 0.35] | [−0.58, 0.67] |
| | | 12 | 633 | 0.41* | 0.26 | 0.12 | −0.37** | −0.65** | −0.86** |
| | | | | [0.06, 0.67] | [−0.09, 0.52] | [−0.18, 0.36] | [−0.53, −0.16] | [−0.97, −0.17] | [−1.42, −0.11] |
| | equities free | 3 | 642 | 0.14 | −0.01 | −0.14 | 0.04 | 0.36** | 0.60** |
| | | | | [−0.24, 0.33] | [−0.50, 0.22] | [−0.61, 0.16] | [−0.01, 0.48] | [0.26, 1.36] | [0.24, 2.05] |
| | | 6 | 639 | 0.15 | 0.13 | 0.05 | −0.19 | −0.28 | −0.37 |
| | | | | [−0.19, 0.35] | [−0.25, 0.34] | [−0.33, 0.25] | [−0.30, 0.10] | [−0.48, 0.47] | [−0.77, 0.89] |
| | | 12 | 633 | 0.09 | 0.15 | 0.12 | −0.32** | −0.60 | −0.88 |
| | | | | [−0.37, 0.44] | [−0.31, 0.43] | [−0.30, 0.45] | [−0.52, −0.07] | [−0.98, 0.02] | [−1.47, 0.19] |

Table H24: Weights Tests: Wald Statistics and Bootstrap p-Values by Lag Length

| | | Three equal | | Yield = dollar | | Equities = rest | |
|---|---|---|---|---|---|---|---|
| Version, start | Lags | U | Inflation | U | Inflation | U | Inflation |
| Nominal, 1996 | 12 | 8.9 (0.984) | 41.9 (0.053) | 6.1 (0.821) | 24.9 (0.024) | 4.0 (0.978) | 18.9 (0.142) |
| Nominal, 1996 | 6 | 6.7 (0.874) | 23.4 (0.035) | 4.0 (0.726) | 13.0 (0.059) | 4.1 (0.670) | 12.0 (0.082) |
| Nominal, 1996 | 3 | 6.8 (0.469) | 19.7 (0.011) | 2.4 (0.579) | 11.2 (0.022) | 5.5 (0.204) | 10.8 (0.031) |
| Real, 1996 | 12 | 7.0 (0.996) | 42.9 (0.041) | 4.1 (0.966) | 22.1 (0.073) | 4.4 (0.960) | 18.9 (0.143) |
| Real, 1996 | 6 | 5.9 (0.912) | 21.6 (0.055) | 3.8 (0.714) | 13.0 (0.048) | 4.1 (0.668) | 10.3 (0.137) |
| Real, 1996 | 3 | 7.9 (0.362) | 18.8 (0.013) | 3.7 (0.339) | 10.0 (0.032) | 5.6 (0.181) | 10.2 (0.033) |
| r*-gap, 1996 | 12 | 21.7 (0.399) | 53.9 (0.013) | 9.9 (0.425) | 22.9 (0.057) | 13.9 (0.216) | 22.6 (0.055) |
| r*-gap, 1996 | 6 | 16.4 (0.236) | 23.6 (0.050) | 11.3 (0.120) | 11.0 (0.094) | 5.2 (0.521) | 11.5 (0.087) |
| r*-gap, 1996 | 3 | 11.6 (0.201) | 17.8 (0.017) | 4.3 (0.302) | 9.0 (0.045) | 5.6 (0.171) | 9.8 (0.040) |
| Nominal, 1973 | 12 | 24.5 (0.930) | 26.0 (0.539) | 9.0 (0.671) | 15.6 (0.242) | 20.0 (0.667) | 9.3 (0.740) |
| Nominal, 1973 | 6 | 11.0 (0.984) | 12.9 (0.495) | 2.7 (0.907) | 7.5 (0.299) | 10.1 (0.785) | 1.9 (0.929) |
| Nominal, 1973 | 3 | 9.6 (0.941) | 13.8 (0.085) | 2.3 (0.643) | 7.8 (0.044) | 9.6 (0.636) | 0.9 (0.868) |
| Real, 1973 | 12 | 22.2 (0.886) | 32.4 (0.232) | 10.7 (0.546) | 16.6 (0.212) | 14.9 (0.718) | 13.3 (0.404) |
| Real, 1973 | 6 | 8.5 (0.990) | 12.3 (0.568) | 4.1 (0.710) | 7.2 (0.358) | 6.2 (0.858) | 3.2 (0.795) |
| Real, 1973 | 3 | 6.9 (0.950) | 8.2 (0.473) | 4.0 (0.358) | 5.6 (0.115) | 5.8 (0.662) | 1.2 (0.800) |
| r*-gap, 1973 | 12 | 33.4 (0.408) | 27.9 (0.374) | 17.6 (0.134) | 11.0 (0.561) | 18.5 (0.358) | 14.4 (0.371) |
| r*-gap, 1973 | 6 | 20.3 (0.622) | 9.2 (0.752) | 11.4 (0.156) | 4.4 (0.626) | 6.9 (0.717) | 3.5 (0.766) |
| r*-gap, 1973 | 3 | 13.2 (0.713) | 5.6 (0.609) | 5.1 (0.272) | 3.5 (0.292) | 6.1 (0.480) | 1.0 (0.800) |

*Notes:* Wald statistic with the bootstrap $p$-value in parentheses; U = unemployment gap equation. Tests as in Table 8.

*Notes:* The daily block is that of the three components in all three versions, so the versions differ only in the monthly equations: one variable (the FCI built from the components), equities separately with the 10-year yield and the dollar as one variable, or all three separately. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

Table H28: Monthly VAR Benchmark (3, 6 and 12 Lags)

| | | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-year | FCI | Identification | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | none | Proxy VAR | 3 | 369 | −0.53 | −0.43* | −0.35* | 0.00 | −0.03 | −0.06 |
| | | | | | [−0.90, 0.06] | [−0.64, −0.10] | [−0.49, −0.13] | [−1.08, 0.77] | [−1.14, 0.76] | [−1.16, 0.72] |
| | | | 6 | 369 | −0.46 | −0.36 | −0.26 | −0.05 | −0.15 | −0.27 |
| | | | | | [−0.93, 0.36] | [−0.76, 0.34] | [−0.57, 0.27] | [−1.54, 0.85] | [−2.09, 1.02] | [−2.34, 0.98] |
| | | | 12 | 369 | −0.91* | −0.61* | −0.32 | 0.03 | −0.08 | −0.26 |
| | | | | | [−1.37, −0.42] | [−1.08, −0.05] | [−0.71, 0.17] | [−1.20, 0.97] | [−1.83, 1.25] | [−2.27, 1.25] |
| | | Window impact | 3 | 369 | −0.61* | −0.46 | −0.36 | 0.14 | 0.12 | 0.09 |
| | | | | | [−0.89, −0.13] | [−0.66, 0.05] | [−0.55, 0.09] | [−0.12, 0.29] | [−0.34, 0.34] | [−0.48, 0.38] |
| | | | 6 | 369 | −0.54* | −0.42 | −0.30 | 0.09 | 0.03 | −0.07 |
| | | | | | [−0.86, −0.16] | [−0.67, 0.05] | [−0.53, 0.11] | [−0.15, 0.27] | [−0.41, 0.31] | [−0.65, 0.31] |
| | | | 12 | 369 | −0.90* | −0.63* | −0.35 | 0.13 | 0.06 | −0.09 |
| | | | | | [−1.42, −0.37] | [−1.01, −0.04] | [−0.74, 0.21] | [−0.11, 0.36] | [−0.41, 0.52] | [−0.73, 0.54] |
| | nominal | Proxy VAR | 3 | 369 | 0.35 | 0.13 | 0.02 | −0.29 | −0.42 | −0.48 |
| | | | | | [−0.34, 173.16] | [−0.40, 100.92] | [−0.38, 62.73] | [−73.88, 0.60] | [−87.91, 0.46] | [−90.42, 0.40] |
| | | | 6 | 369 | 0.63 | 0.37 | 0.17 | −0.49 | −0.78 | −0.95 |

| | | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-year | FCI | Identification | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| | | | | | unbounded | unbounded | unbounded | unbounded | unbounded | unbounded |
| | | | 12 | 369 | 0.72 | 0.51 | 0.32 | −0.67 | −1.02 | −1.17 |
| | | | | | unbounded | unbounded | unbounded | unbounded | unbounded | unbounded |
| | | Window impact | 3 | 369 | −0.06 | 0.02 | 0.00 | −0.06 | −0.22 | −0.32 |
| | | | | | [−0.41, 0.31] | [−0.30, 0.44] | [−0.31, 0.42] | [−0.32, 0.15] | [−0.63, 0.14] | [−0.80, 0.16] |
| | | | 6 | 369 | 0.02 | 0.08 | 0.05 | −0.12 | −0.34* | −0.52* |
| | | | | | [−0.33, 0.39] | [−0.28, 0.42] | [−0.32, 0.35] | [−0.36, 0.04] | [−0.78, −0.03] | [−1.09, −0.06] |
| | | | 12 | 369 | −0.14 | −0.09 | −0.01 | −0.13 | −0.36 | −0.57 |
| | | | | | [−0.64, 0.38] | [−0.60, 0.43] | [−0.51, 0.48] | [−0.38, 0.12] | [−0.86, 0.15] | [−1.29, 0.16] |
| Real | none | Proxy VAR | 3 | 369 | −0.45* | −0.41* | −0.35* | −0.04 | −0.06 | −0.08 |
| | | | | | [−0.88, −0.13] | [−0.65, −0.22] | [−0.52, −0.22] | [−0.68, 0.62] | [−0.72, 0.62] | [−0.73, 0.59] |
| | | | 6 | 369 | −0.48* | −0.38 | −0.26 | −0.06 | −0.17 | −0.31 |
| | | | | | [−1.04, −0.05] | [−0.84, 0.03] | [−0.60, 0.06] | [−0.91, 0.76] | [−1.29, 0.89] | [−1.51, 0.82] |
| | | | 12 | 369 | −0.72* | −0.48* | −0.22 | −0.12 | −0.29 | −0.46 |
| | | | | | [−1.29, −0.35] | [−0.99, −0.07] | [−0.62, 0.15] | [−0.97, 0.71] | [−1.50, 0.85] | [−1.85, 0.80] |
| | | Window impact | 3 | 369 | −0.38* | −0.37* | −0.32 | −0.04 | −0.06 | −0.07 |
| | | | | | [−0.67, −0.05] | [−0.61, −0.01] | [−0.56, 0.01] | [−0.26, 0.14] | [−0.45, 0.23] | [−0.60, 0.29] |
| | | | 6 | 369 | −0.41 | −0.34 | −0.24 | −0.03 | −0.13 | −0.26 |
| | | | | | [−0.73, 0.01] | [−0.64, 0.14] | [−0.47, 0.19] | [−0.32, 0.17] | [−0.67, 0.19] | [−0.98, 0.17] |
| | | | 12 | 369 | −0.65* | −0.46 | −0.23 | −0.04 | −0.16 | −0.29 |
| | | | | | [−1.12, −0.15] | [−0.94, 0.02] | [−0.66, 0.23] | [−0.30, 0.20] | [−0.66, 0.31] | [−0.97, 0.35] |
| | real | Proxy VAR | 3 | 369 | 0.26 | 0.01 | −0.09 | −0.23 | −0.30 | −0.32 |
| | | | | | [−0.27, 2.00] | [−0.40, 1.23] | [−0.38, 0.77] | [−1.28, 0.48] | [−1.48, 0.40] | [−1.54, 0.37] |
| | | | 6 | 369 | 0.49 | 0.26 | 0.07 | −0.46 | −0.72 | −0.85 |
| | | | | | [−0.22, 4.80] | [−0.35, 3.58] | [−0.34, 2.24] | [−3.03, 0.44] | [−4.25, 0.40] | [−4.56, 0.31] |
| | | | 12 | 369 | 0.72 | 0.49 | 0.30 | −0.72 | −1.05 | −1.12 |
| | | | | | [−0.21, 24.27] | [−0.33, 19.69] | [−0.25, 12.47] | [−11.42, 0.16] | [−16.08, 0.22] | [−16.57, 0.33] |
| | | Window impact | 3 | 369 | 0.04 | −0.04 | −0.10 | −0.18 | −0.28 | −0.32 |
| | | | | | [−0.29, 0.38] | [−0.35, 0.36] | [−0.36, 0.28] | [−0.41, 0.04] | [−0.64, 0.10] | [−0.76, 0.16] |
| | | | 6 | 369 | 0.06 | 0.04 | −0.03 | −0.23* | −0.45* | −0.60* |
| | | | | | [−0.37, 0.43] | [−0.36, 0.46] | [−0.37, 0.37] | [−0.55, −0.02] | [−0.97, −0.07] | [−1.22, −0.07] |
| | | | 12 | 369 | −0.02 | −0.04 | 0.00 | −0.26 | −0.49 | −0.63 |
| | | | | | [−0.61, 0.48] | [−0.54, 0.47] | [−0.49, 0.44] | [−0.57, 0.01] | [−1.00, 0.02] | [−1.30, 0.08] |
| r*-gap | none | Proxy VAR | 3 | 369 | −0.27 | −0.22* | −0.17* | −0.10 | −0.13 | −0.15 |
| | | | | | [−0.80, 0.06] | [−0.51, −0.03] | [−0.34, −0.06] | [−0.71, 0.59] | [−0.75, 0.57] | [−0.77, 0.55] |
| | | | 6 | 369 | −0.21 | −0.11 | −0.03 | −0.18 | −0.38 | −0.56 |
| | | | | | [−0.86, 0.18] | [−0.61, 0.22] | [−0.36, 0.21] | [−0.88, 0.67] | [−1.28, 0.68] | [−1.49, 0.54] |
| | | | 12 | 369 | −0.33 | −0.10 | 0.06 | −0.29 | −0.57 | −0.79 |
| | | | | | [−0.98, 0.02] | [−0.64, 0.24] | [−0.30, 0.32] | [−0.97, 0.48] | [−1.51, 0.47] | [−1.83, 0.31] |
| | | Window impact | 3 | 369 | −0.16 | −0.17 | −0.14 | −0.13 | −0.15 | −0.17 |
| | | | | | [−0.56, 0.23] | [−0.52, 0.25] | [−0.46, 0.21] | [−0.37, 0.09] | [−0.58, 0.20] | [−0.71, 0.28] |
| | | | 6 | 369 | −0.06 | −0.02 | 0.03 | −0.24* | −0.45* | −0.61* |
| | | | | | [−0.62, 0.33] | [−0.48, 0.41] | [−0.34, 0.38] | [−0.56, 0.00] | [−1.05, −0.02] | [−1.28, 0.00] |
| | | | 12 | 369 | −0.19 | −0.03 | 0.09 | −0.27* | −0.51* | −0.69* |
| | | | | | [−0.75, 0.23] | [−0.52, 0.37] | [−0.28, 0.44] | [−0.53, 0.00] | [−0.96, −0.02] | [−1.25, −0.05] |
| | r*-gap | Proxy VAR | 3 | 369 | 0.33 | 0.13 | −0.01 | −0.19 | −0.24 | −0.25 |
| | | | | | [−0.15, 1.34] | [−0.26, 0.93] | [−0.30, 0.59] | [−0.89, 0.48] | [−0.95, 0.43] | [−0.98, 0.40] |
| | | | 6 | 369 | 0.67 | 0.41 | 0.16 | −0.59 | −0.87 | −1.00* |
| | | | | | [0.00, 2.51] | [−0.12, 1.84] | [−0.20, 1.11] | [−1.80, 0.12] | [−2.40, 0.02] | [−2.53, −0.09] |
| | | | 12 | 369 | 0.95* | 0.73 | 0.47 | −0.84* | −1.24* | −1.35* |
| | | | | | [0.01, 5.87] | [−0.04, 4.55] | [−0.05, 2.89] | [−3.17, −0.15] | [−4.39, −0.24] | [−4.51, −0.24] |
| | | Window impact | 3 | 369 | 0.19 | 0.06 | −0.05 | −0.19 | −0.24 | −0.26 |

| | | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-year | FCI | Identification | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| | | | | | [−0.26, 0.56] | [−0.39, 0.47] | [−0.46, 0.36] | [−0.43, 0.05] | [−0.65, 0.16] | [−0.79, 0.27] |
| | | | 6 | 369 | 0.36 | 0.22 | 0.04 | −0.41* | −0.67* | −0.81* |
| | | | | | [−0.14, 0.81] | [−0.35, 0.60] | [−0.42, 0.27] | [−0.72, −0.15] | [−1.20, −0.14] | [−1.45, −0.12] |
| | | | 12 | 369 | 0.33 | 0.31 | 0.25 | −0.45* | −0.77* | −0.94* |
| | | | | | [−0.25, 0.82] | [−0.25, 0.76] | [−0.23, 0.60] | [−0.71, −0.16] | [−1.26, −0.20] | [−1.54, −0.18] |

*Notes:* Monthly VAR in the unemployment gap, core PCE inflation, the end-of-month 2-year yield and, when included, the end-of-month FCI (same version), 1996 to September 2026, no trend, nothing dropped. Proxy VAR: Nakamura–Steinsson surprise as instrument; sets are exact 90% Anderson–Rubin sets, "unbounded" when the instrument cannot reject large effects. Window impact: impact vector taken from the daily model (the end-of-month response to a window surprise); 90% bootstrap intervals. Unit in both: +1 pp on the 2-year yield (the end-of-month response to a +1 pp window surprise is 0.92 to 1.01, so the two units are the same up to 8%). * = 90% interval excludes zero.

Table H25: Standard Methods with a Commodity Price Index in the First Block

| | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|
| Method | Sample | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Cholesky + commodity prices, 2-year yield | 1996–2026 | 369 | −0.81* | −0.44 | −0.13 | 0.11 | 0.06 | −0.05 |
| | | | [−1.24, −0.28] | [−0.83, 0.03] | [−0.45, 0.34] | [−0.12, 0.35] | [−0.36, 0.48] | [−0.64, 0.49] |
| Cholesky + commodity prices, federal funds rate | 1984–2007 | 288 | −0.03 | −0.02 | −0.01 | −0.02 | −0.04 | −0.04 |
| | | | [−0.07, 0.01] | [−0.06, 0.03] | [−0.04, 0.04] | [−0.09, 0.02] | [−0.13, 0.05] | [−0.15, 0.07] |

*Notes:* Monthly data, 12 lags, constant, no trend; unemployment gap, core PCE inflation and the end-of-month policy rate (the 2-year Treasury yield, or the effective federal funds rate where stated), policy rate ordered last in the Cholesky VARs; price level = cumulated inflation/12. The log PPI for all commodities (monthly average) is added to the system and ordered before the policy rate. Unit: +1 pp on the policy rate on impact. Brackets: 90% intervals (residual bootstrap for the Cholesky VARs; exact Anderson–Rubin sets for the proxy VAR). * = excludes zero. Specification: baseline/lib/litstyle.py.

Table H26: Monthly Proxy VAR: First-Stage Strength and Impact on the FCI

| 2-year | FCI | $F$ (3 lags) | $F$ (6) | $F$ (12) | Proxy impact on FCI | Model impact on FCI |
|---|---|---|---|---|---|---|
| Nominal | none | 5.0 | 4.6 | 4.5 | | |
| | nominal | 2.7 | 2.1 | 1.6 | 1.73 | 1.13 |
| Real | none | 9.0 | 7.3 | 6.9 | | |
| | real | 6.0 | 4.2 | 3.0 | 1.50 | 1.03 |
| r*-gap | none | 9.1 | 8.2 | 8.0 | | |
| | r*-gap | 8.4 | 6.7 | 4.4 | 1.44 | 1.03 |

*Notes:* $F$: robust first-stage statistic of the policy-variable residual on the instrument (instrument months only, 1996–2026). Impact on FCI: end-of-month response of the FCI per +1 pp on the end-of-month 2-year yield, estimated from monthly residuals (proxy VAR, 12 lags) and implied by the daily model.

Table H27: Monthly VAR Benchmark (12 Lags)

| | | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-year | FCI | Identification | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | none | Proxy VAR | 12 | 369 | −0.91* | −0.61* | −0.32 | 0.03 | −0.08 | −0.26 |
| | | | | | [−1.37, −0.42] | [−1.08, −0.05] | [−0.71, 0.17] | [−1.20, 0.97] | [−1.83, 1.25] | [−2.27, 1.25] |
| | | Window impact | 12 | 369 | −0.90* | −0.63* | −0.35 | 0.13 | 0.06 | −0.09 |
| | | | | | [−1.42, −0.37] | [−1.01, −0.04] | [−0.74, 0.21] | [−0.11, 0.36] | [−0.41, 0.52] | [−0.73, 0.54] |
| | nominal | Proxy VAR | 12 | 369 | 0.72 | 0.51 | 0.32 | −0.67 | −1.02 | −1.17 |
| | | | | | unbounded | unbounded | unbounded | unbounded | unbounded | unbounded |
| | | Window impact | 12 | 369 | −0.14 | −0.09 | −0.01 | −0.13 | −0.36 | −0.57 |
| | | | | | [−0.64, 0.38] | [−0.60, 0.43] | [−0.51, 0.48] | [−0.38, 0.12] | [−0.86, 0.15] | [−1.29, 0.16] |
| Real | none | Proxy VAR | 12 | 369 | −0.72* | −0.48* | −0.22 | −0.12 | −0.29 | −0.46 |
| | | | | | [−1.29, −0.35] | [−0.99, −0.07] | [−0.62, 0.15] | [−0.97, 0.71] | [−1.50, 0.85] | [−1.85, 0.80] |
| | | Window impact | 12 | 369 | −0.65* | −0.46 | −0.23 | −0.04 | −0.16 | −0.29 |
| | | | | | [−1.12, −0.15] | [−0.94, 0.02] | [−0.66, 0.23] | [−0.30, 0.20] | [−0.66, 0.31] | [−0.97, 0.35] |
| | real | Proxy VAR | 12 | 369 | 0.72 | 0.49 | 0.30 | −0.72 | −1.05 | −1.12 |
| | | | | | [−0.21, 24.27] | [−0.33, 19.69] | [−0.25, 12.47] | [−11.42, 0.16] | [−16.08, 0.22] | [−16.57, 0.33] |
| | | Window impact | 12 | 369 | −0.02 | −0.04 | 0.00 | −0.26 | −0.49 | −0.63 |
| | | | | | [−0.61, 0.48] | [−0.54, 0.47] | [−0.49, 0.44] | [−0.57, 0.01] | [−1.00, 0.02] | [−1.30, 0.08] |
| r*-gap | none | Proxy VAR | 12 | 369 | −0.33 | −0.10 | 0.06 | −0.29 | −0.57 | −0.79 |
| | | | | | [−0.98, 0.02] | [−0.64, 0.24] | [−0.30, 0.32] | [−0.97, 0.48] | [−1.51, 0.47] | [−1.83, 0.31] |
| | | Window impact | 12 | 369 | −0.19 | −0.03 | 0.09 | −0.27* | −0.51* | −0.69* |
| | | | | | [−0.75, 0.23] | [−0.52, 0.37] | [−0.28, 0.44] | [−0.53, 0.00] | [−0.96, −0.02] | [−1.25, −0.05] |
| | r*-gap | Proxy VAR | 12 | 369 | 0.95* | 0.73 | 0.47 | −0.84* | −1.24* | −1.35* |
| | | | | | [0.01, 5.87] | [−0.04, 4.55] | [−0.05, 2.89] | [−3.17, −0.15] | [−4.39, −0.24] | [−4.51, −0.24] |
| | | Window impact | 12 | 369 | 0.33 | 0.31 | 0.25 | −0.45* | −0.77* | −0.94* |
| | | | | | [−0.25, 0.82] | [−0.25, 0.76] | [−0.23, 0.60] | [−0.71, −0.16] | [−1.26, −0.20] | [−1.54, −0.18] |

*Notes:* Monthly VAR in the unemployment gap, core PCE inflation, the end-of-month 2-year yield and, when included, the end-of-month FCI (same version), 1996 to September 2026, no trend, nothing dropped. Proxy VAR: Nakamura–Steinsson surprise as instrument; sets are exact 90% Anderson–Rubin sets, "unbounded" when the instrument cannot reject large effects. Window impact: impact vector taken from the daily model (the end-of-month response to a window surprise); 90% bootstrap intervals. Unit in both: +1 pp on the 2-year yield (the end-of-month response to a +1 pp window surprise is 0.92 to 1.01, so the two units are the same up to 8%). * = 90% interval excludes zero.

Table H29: A Monthly Proxy VAR when the r*-gap Model is the True Process

| Estimate | Data | True | Simulated median | 5th–95th percentile |
|---|---|---|---|---|
| *A. Proxy VAR with the FCI* | | | | |
| First-stage $F$ | 4.4 | | 11.1 | [2.5, 27.2] |
| Unemployment gap, 12 m | 0.95 | 0.33 | 0.35 | [−0.31, 1.26] |
| Price level, 24 m | −1.24 | −0.80 | −0.69 | [−1.91, 0.17] |
| *B. Proxy VAR without the FCI* | | | | |
| Unemployment gap, 12 m | −0.33 | 0.33 | 0.02 | [−0.70, 0.70] |
| *C. Monthly VAR with the window impact and the FCI* | | | | |
| Unemployment gap, 12 m | 0.33 | 0.34 | 0.35 | [−0.10, 0.80] |

*Notes:* 1,000 samples of 369 months generated by the estimated r*-gap model with the FCI (12 lags), with the data's calendar and announcement days; policy reaches unemployment and prices only through the 2-year yield and the FCI. Data: the 1996 sample, 12 lags, r*-gap measures; end-of-month series in the monthly VAR. True: the model's response per +1 pp on the 2-year yield, in the window (panel C) and on the end-of-month yield (panels A and B; the window response divided by the model's end-of-month response of 0.92–1.01). Simulated: median and 5th–95th percentiles of the estimates.

# Appendix I Construction and Estimation Details

**Events.** I use the 252 FOMC announcements from January 1996 to July 2026 with all the window measures available in the U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). As in the companion note, I leave out unscheduled announcements with no change in the policy rate, and March–April 2020. The announcements of September and October 2026 are left out because too few later days are available. The DXY index used in the window covers fewer currencies than the broad dollar used in the daily and monthly index, which I treat as a measurement difference.

**From monthly to daily.** Expected inflation and $r^*$ are not available daily, and they move slowly. I take the monthly gap between $R_t^v$ in equation (13) and $y_t^2$, interpolate it linearly between the middles of adjacent months, and add it to the daily yield. The daily index is

$$F_\tau^v = w_B\, y_\tau^{10} + w_Q\, 100 \ln \mathrm{CAPE}_\tau + w_D\, 100 \ln \mathrm{DOL}_\tau + o_\tau^v, \tag{18}$$

where $\mathrm{DOL}_\tau$ is the daily nominal broad dollar. The offset $o_\tau^v$ is the monthly difference between the index in equation (9) and the average of the first three terms, interpolated in the same way. It also absorbs the difference between the real and the nominal dollar. The monthly averages of the daily series are very close to the monthly measures: the root-mean-square difference is at most 0.02 unit for the FCI and 0.04 percentage point for the two-year yield. The offsets change little from day to day, so day-to-day moves in the index come from market prices. Table I30 shows the measures used at each frequency. The daily calendar has every weekday; missing values, such as on holidays, are filled from the previous observation for up to three days. Each daily equation has 13 coefficients with the FCI; each monthly equation has $1 + 4L$ with the FCI and $1 + 3L$ without it.

**Tracing the effect of a surprise.** The response is computed recursively, as a deviation from a path with no surprise. Let the announcement occur on weekday $d_0$ of the announcement month, and let $h = 0, 1, \ldots, 36$ count months from then. Tildes denote responses.

1. In each month, unemployment and inflation come first: $\tilde{m}_0 = 0$, and for $h \geq 1$,

$$\tilde{m}_h = \sum_{l=1}^{L} A_l\, \tilde{m}_{h-l} + \sum_{l=1}^{L} G_l\, \tilde{x}_{h-l}^{M}, \tag{19}$$

   with terms dated before the announcement month equal to zero, and $\tilde{x}_h^M$ the response of the monthly average of $x$.

2. The days of the month follow. Before day $d_0$, $\tilde{x}_\tau = 0$. On day $d_0$, $\tilde{x}_\tau = b$ from equation (2). On every later day, with $\tilde{x}_{\tau-1}^{(k)}$ the average of the responses over the $k$ days through $\tau - 1$,

$$\tilde{x}_\tau = \sum_{k \in \{1,5,22,66\}} \Phi_k\, \tilde{x}_{\tau-1}^{(k)} + \Gamma_0\, \tilde{m}_{h(\tau)} + \Gamma_1\, \tilde{m}_{h(\tau)-1}, \tag{20}$$

   where $h(\tau)$ is the month of day $\tau$, counted from the announcement month.

Table I30: Measures of Stocks and the Dollar at Each Frequency

| Frequency | Stocks | Dollar |
|---|---|---|
| Window | S&P 500 | DXY index |
| Daily | CAPE | Nominal broad dollar, plus offset |
| Monthly | CAPE, or earnings-yield gap to $r^*$ | Real broad dollar |

3. The monthly average $\tilde{x}_h^M$ then enters equation (19) for later months.

Months after the announcement month have 21 weekdays. To avoid computing a path for every date, I form six groups of announcement dates by the weekday of the month, in blocks of four days, and compute the path at each group's average day and month length. With $\omega_g$ the share of announcements in group $g$, the reported response is $\sum_g \omega_g \tilde{m}_h^{(g)}$. Because the model is linear, these responses do not depend on the constant or the trend.

**Three-component specification.** Each component's equation in the daily block contains its own 1-, 5-, 22- and 66-day averages and those of the two-year yield, plus the monthly macro variables, and the two-year yield's equation its own averages and the macro variables. That is 48 coefficients in all, against 26 in the baseline daily block with the FCI. The three window impacts add up to the impact of the FCI.

**Ordinary movement of the FCI.** The ordinary loading is the covariance of the FCI and yield innovations divided by the variance of the yield innovation, in FCI units per percentage point on the yield, so it is comparable with the window loading $b_F$. For path B of Section 3.4 I take the innovations from the daily block, the nearest ordinary observations to the announcement window: the residuals of equation (3) on days without an announcement (conditioning on the 1-, 5-, 22- and 66-day averages of the yield and the FCI and on the monthly unemployment gap and inflation of this month and last). The 90 percent interval re-estimates the daily block and the loading on the days of each block-bootstrap draw. The earlier estimate in monthly data is the same ratio from the residuals of a monthly VAR in unemployment, inflation, the end-of-month two-year yield and the end-of-month FCI of the same version, with 12 lags, over 1996 to 2026. Neither measures an announcement. Daily innovations are close-to-close changes that include all of the day's news; monthly innovations add the dynamics within the month. Because the model is linear, the response to any initial FCI movement $\phi$ is the response to the yield alone plus $\phi$ times the response to the FCI, so path B can be recomputed for any loading: with the monthly loading (0.51) the A − B contrast after 24 months is 0.31 for unemployment and −0.22 for the price level, and with no FCI movement at all (a loading of zero) 0.65 and −0.46.

**Simulation.** The process is the estimated baseline model (relative to $r^*$, with the FCI; 12 lags), on the data's calendar of weekdays from January 1996 to September 2026 (369 months) and the actual announcement days. Within each month the macro variables are set first and the days of the month follow. Daily innovations are drawn with replacement from the model's daily residuals, separately for announcement days and other days; on announcement days the policy impact $b\,s_a$ is first removed from the residual and then added back with a drawn surprise. Monthly innovations

are drawn jointly from the monthly residuals. Surprises are drawn with replacement from the 252 observed surprises. The twelve months before the sample and the 70 days before it are the actual 1995 data. On each sample I estimate the proxy VAR with and without the FCI, with the instrument and instrument months as in the data, and the window-impact VAR, taking the impact on the yield and the FCI from the model.

**Bootstrap.** The draws are those of Appendix D: 500 block-bootstrap draws, from each of which I compute the response and report percentile intervals. The sets for the Phillips multiplier in Table 6 use the same draws.

## Appendix J Data Sources

Table J31 lists the data. Series from FRED are identified by their mnemonics. Quarterly series are placed in the middle month of the quarter and interpolated linearly.

Table J31: Data Sources

| Series | Frequency | Source |
|---|---|---|
| Unemployment rate | Monthly | FRED UNRATE |
| Natural rate of unemployment | Quarterly | CBO, FRED NROU |
| Core PCE price index | Monthly | FRED PCEPILFE |
| Two-year Treasury yield | Daily | FRED DGS2; before June 1976, the Gürkaynak–Sack–Wright zero-coupon yield (Gürkaynak, Sack and Wright, 2007) (Federal Reserve Board) |
| 10-year Treasury yield | Daily, monthly | FRED DGS10, GS10 |
| Expected inflation, 1, 2 and 10 years | Monthly | Cleveland Fed, FRED EXPINF1YR, EXPINF2YR, EXPINF10YR; before 1982, Livingston Survey (Philadelphia Fed) |
| $r^*$, Laubach–Williams | Quarterly | New York Fed, one-sided estimates (data through 2026Q2) |
| $r^*$, Kiley | Quarterly | One-sided estimates of the author |
| CAPE | Monthly | Shiller, spliced over 1990–92 to the monthly average of daily CAPE |
| Daily CAPE | Daily | S&P 500 (FRED SP500) over Shiller's 10-year earnings, constructed by the daily workflow of the repository |
| Real broad dollar | Monthly | FRED RTWEXBGS, spliced to TWEXBPA in 2006 |
| Nominal broad dollar | Daily | FRED DTWEXBGS; earlier DTWEXB, TWEXBMTH |
| Excess bond premium | Monthly | Gilchrist and Zakrajšek (2012), Federal Reserve Board |
| Federal funds rate; commodity prices | Daily; monthly | FRED DFF; FRED PPIACO (producer price index, all commodities) |
| Futures, yields, S&P 500 and DXY in the window | Event | U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025) |
| Orthogonalized surprise | Event | Bauer and Swanson (2023*b*), posted data (announcements through 2019) |